Performance of Rust language

May 2026

About us

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- Developer and fan of system SW (compilers, runtimes, verification tools, etc.)
- Compiler teamlead

Zakhar Akimov, https://github.com/z-inform

- Developer of general-purpose and AI compilers
- ML-in-compilers expert

Rust language features

- Language for high-performance system programming
  - Zero cost/overhead abstractions: don’t pay for what you don’t use + no (or minimal) runtime overhead (iterators, closures, traits, etc.)
  - Allows low-level tuning (SIMD, inline assembler, compiler intrinsics like __builtin_expect, etc.)
- Supports modern programming paradigms
  - Procedural, functional and object-oriented programming
  - Static and runtime polymorphism
  - Metaprogramming
- Reduces number of UBs through stricter language rules and runtime checks
  - Especially memory safety

Rust performance: Disadvantages

- Runtime checks
- Forced initialization
- Integer overflows are defined behavior
- Performance-robustness tradeoffs in stdlib
- …

Rust performance: Advantages

- More aggressive alias analysis
  - Basically restrict-by-default
- Structure layout optimizations
  - Field reorder and niche optimizations
- More efficient standard library containers
- ...

Goals of this talk

https://pxhere.com/ru/photo/870592

- How “expensive” are Rust checks in practice?
  - Compared to Rust without checks and/or C/C++
  - What are the immediate (extra instructions) and indirect (disabling optimizations, I$/BTB pressure) consequences
- Can Sufficiently Smart (Heroic) Compilers © help to mitigate overheads?
- And if not, can developer help compiler to get rid of them?
- On the contrary, what language features help compiler to optimize code better?

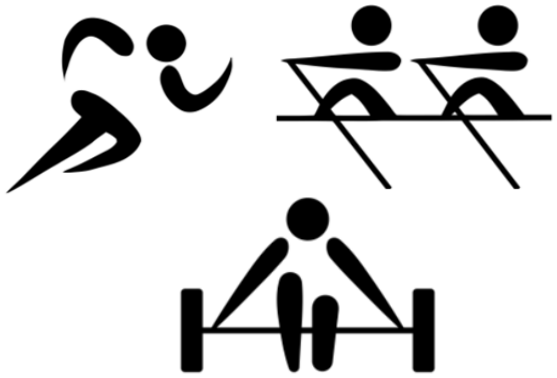
Opinion of language creators

- Steven Klabnik:
  - "Generally Rust is the same order of magnitude as C. Sometimes faster, sometimes slower”
  - “Rust prioritizes memory safety above all else. But speed is a close second"

Overhead analysis

- How can we estimate overheads caused by checks?
- Rust compiler does not provide flags to disable safety checks
- We will compare regular Rust with patched Rust with disabled checks
  - Repo with modified compiler: yugr/rust-private
  - Baseline branch yugr/baseline (on top of 1.87.0)
    - Disabled hashtable randomization and forced function alignment for better measurement stability

Benchmarks



- Rust compiler runtime benchmarks
- Benchmarks from large open-source Rust projects from various domains:
  - Gamedev (Bevy, Veloren)
  - Databases (SpacetimeDB)
  - Text processing (Zed, Ruff, regex)
  - Codecs (rav1e, oxipng)
  - Linear algebra (nalgebra)
  - Asynchronous runtime (tokio)
  - Packet managers (uv)
- Many benchmarks have already been optimized (via unsafe, etc.)

Akovin, https://commons.wikimedia.org/wiki/File:Olympic_types_of_sport_with_icons.png

Rust and C++

- Rust and C++ are similar in expressive power:
  - Support several programming paradigms (procedural, OOP, functional)
  - Different types of polymorphism
  - Etc.
- Similar application domain (high-performance system programming)
  - Zero cost/overhead abstractions
  - Support for low-level tuning
- Same backend optimizers (LLVM, GCC) and optimization technology (PGO, LTO, etc.)
- “Safe” C++ dialect (Hardened C/C++) has similar runtime checks

Not covered: Other language aspects

- Ergonomics, expressiveness (e.g. self-referential data structures), safety, etc.
- A lot of talks on these topics already!
  - Rust Features that I Want in C++ (David Sankel, CppNow 2022)
  - The existential threat against C++ and where to go from here (Helge Penne, NDC 2024)
  - Rust vs C++ with Steve Klabnik and Herb Sutter (SW Engineering Daily podcast)

Not covered: Parallel programming

- Easy and reliable parallel programming is considered one of the big advantages of Rust
  - Fearless Concurrency with Rust
- But this requires a separate talk and deeper expertise
  - RustCon 2027 anyone?

Not covered: Unsafe code

- Rust language consists of two parts:
  - Safe code (most of Rust code)
  - Unsafe code:
    - "Things the Rust authors will implore you not to do" (Rustonomicon)
    - "Aspects of the language you may not use every day and useful in very specific situations" (Rust Book)
    - Used for hotspot optimizations
    - Used for tuning in real-world projects (including benchmarks that we consider)
- In this talk we focus on *safe* code optimizations:
  - (Also not covering third-party libraries, SIMD and compiler intrinsics)
  - Unsafe constructs remove the main advantage of the language
  - Unsafe optimizations are well-known and often trivial

Tricia J, https://www.flickr.com/photos/cazatoma/8746664127

Not covered: Non-idiomatic code

- Nikita Popov, LLVM lead maintainer:
  - “Languages like C, C++ and Rust are by-design on equal footing“
  - “You can only compare how easy it is to write performant code or how performant idiomatic code is”
- Scott McMurray, Rust compiler team:
  - “Clang and rustc are both producing LLVM IR and running mostly the same LLVM optimization passes”
  - “Given enough work they can always produce essentially the same thing, making the comparisons pointless”

Bounds checks

Detailed analysis

Buffer overflows

- Morris Worm (1988)
- Writing excessively large volume of data into program variable

```
char local_buf[32];
sprintf(buf, “Message from user: %s”, received_data);
```

- Stack overflow
  - Stack Smashing, Return-to-libc, Return-Oriented Programming attacks
  - The most standardized and widely used type of attacks
- Heap overflow

https://www.rawpixel.com/image/5958324/free-public-domain-cc0-photo

Prevalence of buffer overflow vulnerabilities

- Leads ratings of the most dangerous vulnerabilities
  - Mitre CWE Top 25 2024 (2, 6, 20 places)
- 70% vulnerabilities in Microsoft products and Chromium caused by memory errors
- And up to 80% memory errors are caused buffer overflow
  - CVE and KEV statistics (2024)
  - Google Project Zero (2024)

Memory safety in Rust

- Frontend generates bounds checks for standard index accesses
  - Arrays, slices, containers, etc.
  - Checks may be eliminated later if optimizer can guarantee correctness
- Uses safer algorithms in stdlib
  - E.g. sorting won't crash on incorrect comparators

Nick Youngson, https://www.picpedia.org/highway-signs/s/safety.html

Overheads

- Comparison and non-taken conditional jump (1-2 ALU slots)
- Extra register for keeping object length for comparison
- I$ pressure (cache misses) due to checking code and panic handlers
  - Branch predictor pressure for never-taken branches is likely none
- Disabling other optimizations
  - Mainly vectorizer

Compiler optimizations

- In many cases compiler can
  - Prove correctness of memory access and remove checks
  - Or extract checks from loops (O(N) -> O(1))
- Such optimizations are done in LLVM optimizer
  - InstCombine, SimpleLoopUnswitch, LoopVectorize, etc.
  - Also help Hardened C/C++
- Optimizer has been significantly improved and many older issues are successfully optimized in modern Rust
  - E.g. well-known examples of Alex Kladov (matklad)

Successful optimization

All checks moved to function prologue

```rust
pub fn foo(a: &[i32], b: &[i32]) -> i32 {
    let mut ans = 0;
    let n = a.len();
    for i in 0..n {
        ans += a[i] * b[i];
    }
    ans
}
```

https://godbolt.org/z/aY744zdfe

Compiled with -Cno-vectorize-loops -Cno-vectorize-slp -Cllvm-args=-unroll-allow-remainder=0

```
foo:
        .cfi_startproc
        test    rsi, rsi
        je      .LBB0_1  // n == 0 => return 0
        lea     rax, [rsi - 1]
        cmp     rcx, rax
        jbe     .LBB0_6  // b.len() < n => panic
        xor     eax, eax
        xor     ecx, ecx
        .p2align        4
.LBB0_4:                 // Loop (without checks!)
        mov     r8d, dword ptr [rdx + 4*rcx]
        imul    r8d, dword ptr [rdi + 4*rcx]
        lea     r9, [rcx + 1]
        add     eax, r8d
        mov     rcx, r9
        cmp     rsi, r9
        jne     .LBB0_4
...
```

Unsuccessful optimization

Checks remain within the loop

```rust
pub fn foo(v: &[i32]) -> i32 {
    let mut ans = 0;
    let chunk_len = 4;
    let num_chunks = v.len() / chunk_len;
    for i in 0..num_chunks {
        for j in 0..chunk_len {
            ans += v[i * chunk_len + j];
        }
    }
    ans
}
```

```
bar:
        ...
        .p2align        4
.LBB0_2:                // Loop (with checks!)
        lea     rcx, [r8 - 3]
        cmp     rcx, rsi
        jae     .LBB0_7
        lea     rcx, [r8 - 2]
        cmp     rcx, rsi
        jae     .LBB0_7
        lea     rcx, [r8 - 1]
        cmp     rcx, rsi
        jae     .LBB0_7
        cmp     r8, rsi
        jae     .LBB0_6
        add     eax, dword ptr [rdi + 4*r8 - 12]
        add     eax, dword ptr [rdi + 4*r8 - 8]
        add     eax, dword ptr [rdi + 4*r8 - 4]
        add     eax, dword ptr [rdi + 4*r8]
        add     r8, 4
        dec     rdx
        jne     .LBB0_2
```

https://godbolt.org/z/176s9nEfE

Compiled with -Cno-vectorize-loops -Cno-vectorize-slp -Cllvm-args=-unroll-allow-remainder=0

Problem with optimizer: delayed panics

- Rust RFC 560 suggested that compiler could combine several checks together
  - If it helps optimization
  - And observed program behavior won't change (modulo error message)
- Similar to classical concept of Imprecise Exceptions
- This has not been decided and optimization currently isn't enabled for bounds checks ☹
- Example: rust #50759

```
use std::fs::{read, write};

fn main() -> Result<(), std::io::Error> {
    let h = read("/tmp/test")?;
    // Each access checked separately ☹
    // (would be enough to check only h[15])
    let b = [h[0], h[1], ... h[15]];
    write("/tmp/test", &b)?;
    Ok(())
}
```

Workarounds: Overview

- Use slices (rather than containers like Vec) in loops
  - This simplifies job of alias analysis
  - Example: fschutt/fastblur #3
- Avoid complex index computations
  - Anything more complicated than adding a constant or adding two variables
  - Even affine operations may break optimizer: Taking Advantage of Auto-Vectorization in Rust
- Do not rely on optimizer in non-trivial cases
  - Optimizations vary (non-monotonically) across compiler versions
  - "Auto-vectorization is not a programming model"
- Same advices are relevant for Hardened C/C++!

Workarounds

- Checks elimination
- Reducing number of checks
- Reducing cost of checks

Workarounds

- **Checks elimination**
- Reducing number of checks
- Reducing cost of checks

Workarounds: Unsafe-code

- The simplest and most robust solution
- Used in stdlib containers and iterators
- Several variants:
  - Explicit pointers in unsafe blocks
  - Check-less APIs like get_unchecked, etc.
  - Compiler hints std::hint::unreachable_unchecked and std::hint::assert_unchecked

```
fn drop(&mut self) {
    // fill the hole again
    unsafe {
        let pos = self.pos;
        ptr::write(
            self.data.get_unchecked_mut(pos),
            self.elt.take().unwrap());
    }
}
```

rust #36072

https://workinjurysource.com/preventing-slip-and-fall-injuries-at-work/

Workarounds: Bounded types

- Index range can be expressed via type system:
  - Enums or bool
- Example: [rust #102303]

```rust
pub fn foo(array: &[i16; 3],
           i: usize) -> i16 {
    array[i]
}
```

```rust
pub enum Enum {
    A, B, C
}

pub fn foo(array: &[i16; 3],
           i: Enum) -> i16 {
    array[i as usize]
}
```

Workarounds

- Checks elimination
- **Reducing number of checks**
- Reducing cost of checks

Workarounds: Manual asserts

- Replaces many checks with one
- Example: jpeg-decoder #167

```
// optimizer hint to eliminate bounds checks
// within loops
assert!(coefficients.len() == 64);

let mut temp = [Wrapping(0); 64];

// columns
for i in 0..8 {
    if coefficients[i + 8] == 0
        && coefficients[i + 16] == 0
        && coefficients[i + 24] == 0
        && coefficients[i + 32] == 0
        && coefficients[i + 40] == 0
        && coefficients[i + 48] == 0
        && coefficients[i + 56] == 0
    {
        ...
```

Workarounds: Reslicing

- Reslicing, preslicing or subslicing
- Pre-construct slices of known size
- Forces compiler to perform *single* check when creating slice
- Example: [dropbox/rust-brotli]

```
for i in 0..n {
    black_box(vec[i]);
}
```

```
let slice = &vec[0..n];
for i in 0..n {
    black_box(slice[i]);
}
```

Lalada, https://www.pexels.com/photo/close-up-of-a-person-cutting-bread-19359855/

Workarounds: Reslicing

- Offsetted slices should better be constructed in two steps to avoid potential integer overflow
- Example: Optimizing rav1d, an AV1 Decoder in Rust

```
&vec[i..(i + n)]
```

```
&vec[i..][..n]
```

Workarounds: Limiting iteration range

- When iterating over several containers explicitly compute common safe range

```rust
pub fn foo(x: &mut [i32], y: &[i32]) {
    let n = x.len();
    for i in 0..n {
        x[i] ^= y[i];
    }
}
```

```rust
pub fn foo(x: &mut [i32], y: &[i32]) {
    let n = std::cmp::min(x.len(), y.len());
    for i in 0..n {
        x[i] ^= y[i];
    }
}
```

https://godbolt.org/z/P8de874c1

```
.LBB1_2:
  cmp rcx, rax
  je  .LBB1_5
  mov r8d, dword ptr [rdx + 4*rax]
  xor dword ptr [rdi + 4*rax], r8d
  lea r8, [rax + 1]
  mov rax, r8
  cmp rsi, r8
  jne .LBB1_2
```

https://godbolt.org/z/6TzEPaaEs

```
.LBB0_2:
  mov     ecx, dword ptr [rdx + 4*rax]
  xor     dword ptr [rdi + 4*rax], ecx
  lea     rcx, [rax + 1]
  mov     rax, rcx
  cmp     rsi, rcx
  jne     .LBB0_2
```

Compiled with
-Cno-vectorize-loops -Cllvm-args=-unroll-allow-remainder=0

Workarounds: Iterators

- Many iterators know container size and can utilize this information to remove bounds checks

```
pub fn foo(x: &mut [i32], y: &[i32]) {
    let n = x.len();
    for i in 0..n {
        x[i] ^= y[i];
    }
}
```

https://godbolt.org/z/P8de874c1

```
.LBB1_2:
  cmp rcx, rax
  je  .LBB1_5
  mov r8d, dword ptr [rdx + 4*rax]
  xor dword ptr [rdi + 4*rax], r8d
  lea r8, [rax + 1]
  mov rax, r8
  cmp rsi, r8
  jne .LBB1_2
```

```
pub fn foo(x: &mut [i32], y: &[i32]) {
    // This also works:
    //   for (x, y) in x.iter_mut().zip(y)
    x.iter_mut()
        .zip(y)
        .for_each(|(x, y)| *x ^= *y);
}
```

https://godbolt.org/z/je69173xv

```
.LBB0_2:
  mov     ecx, dword ptr [rdx + 4*rax]
  xor     dword ptr [rdi + 4*rax], ecx
  lea     rcx, [rax + 1]
  mov     rax, rcx
  cmp     rsi, rcx
  jne     .LBB0_2
```

Compiled with
-Cno-vectorize-loops -Cllvm-args=-unroll-allow-remainder=0

Disadvantages of iterators

- “Miracle of zero cost abstraction has a limit" (fridsun at Reddit)
- Significant slowdown and code size increase for debug builds
- Compiler sometimes fails to inline long iterator chains
- Iterator lambdas take references to container elements which may cause issues with alias analysis (rust #106539)
- Chained iterators (chain, flat_map, flatten, ..=, etc.) may generate inefficient code in for-loops (“external iteration”)
  - Use only with .for_each, .fold, .find, etc. (“internal iteration”)
  - Are iterators even efficient?

Karen Arnolds, https://www.publicdomainpictures.net/en/view-image.php?image=235675&picture=man-carrying-stack-of-boxes

Workarounds: Order of accesses

- Prefer to access data starting with farthest element

```
let a = x[0];
let b = x[1];
let c = x[2];
```

```
let c = x[2];
let b = x[1];
let a = x[0];
```

or

```
let a = x[0];
let b = x[1];
let c = x[2];
```

```
let [a, b, c] = data[..3] else {
    panic!()
};
```

or

```
let a = x[0];
let b = x[1];
let c = x[2];
```

```
let [a, b, c] = data[..3]
    .try_into().unwrap();
```

Workarounds: SIMD

- Developer can manually vectorize code
- Stdlib provides
  - Portable std::simd API (nightly only)
  - Platform vector intrinsics std::arch (unsafe)
- There are also many third-party libraries

```rust
pub fn sum_simd(data: &[i32]) -> i32 {
    const LANES: usize = 8;  // AVX

    let mut acc = i32x8::splat(0);

    for chunk in data.chunks_exact(LANES) {
        let v = i32x8::from_slice(chunk);
        acc += v;
    }

    let mut tail_acc = 0i32;
    for &x in data.chunks_exact(LANES).remainder() {
        tail_acc += x;
    }

    acc.reduce_sum() + tail_acc
}
```

https://godbolt.org/z/G3bfGEhnG

Workarounds

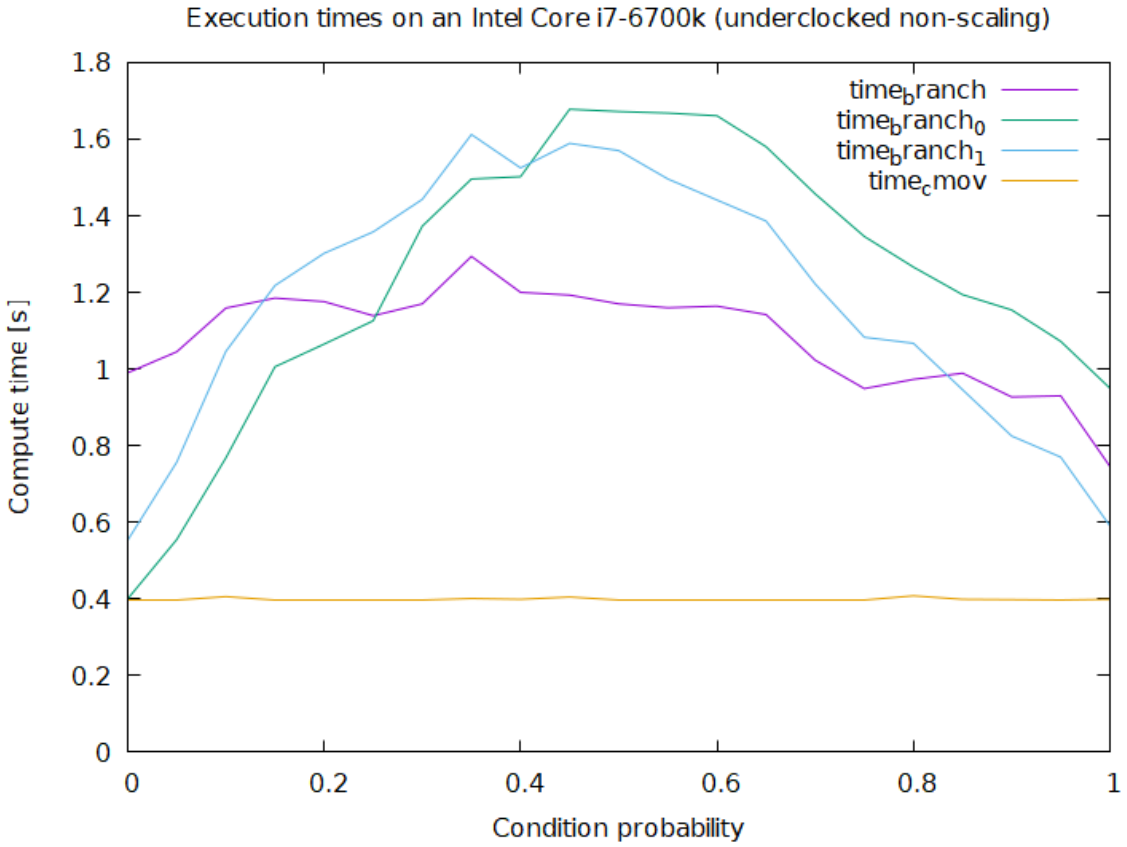
- Checks elimination
- Reducing number of checks
- **Reducing cost of checks**

Workarounds: Arithmetic instead of checks

- Checks may be replaced with conditional move

```
let i = cmp::min(i, sums.len() - 1);
sums[i] = ...
```

- CMOV may be faster or slower than branch
- Disadvantage: bounds errors will now be ignored



https://github.com/marcin-osowski/cmov

Workarounds: Arithmetic instead of checks

- Bounds checks may be replaced with masking if length is 2^N (or can be padded to 2^N):

```
const SIZE: usize = 100;
const LOOKUP_TABLE_SIZE: usize = SIZE.next_power_of_two();

let table: [u64; LOOKUP_TABLE_SIZE] = ...
... table[table & (LOOKUP_TABLE_SIZE - 1)] ...
```

- AND is *usually* faster (e.g. can run on more ALUs in processor)
- Unfortunately optimization (currently) applicable only to arrays

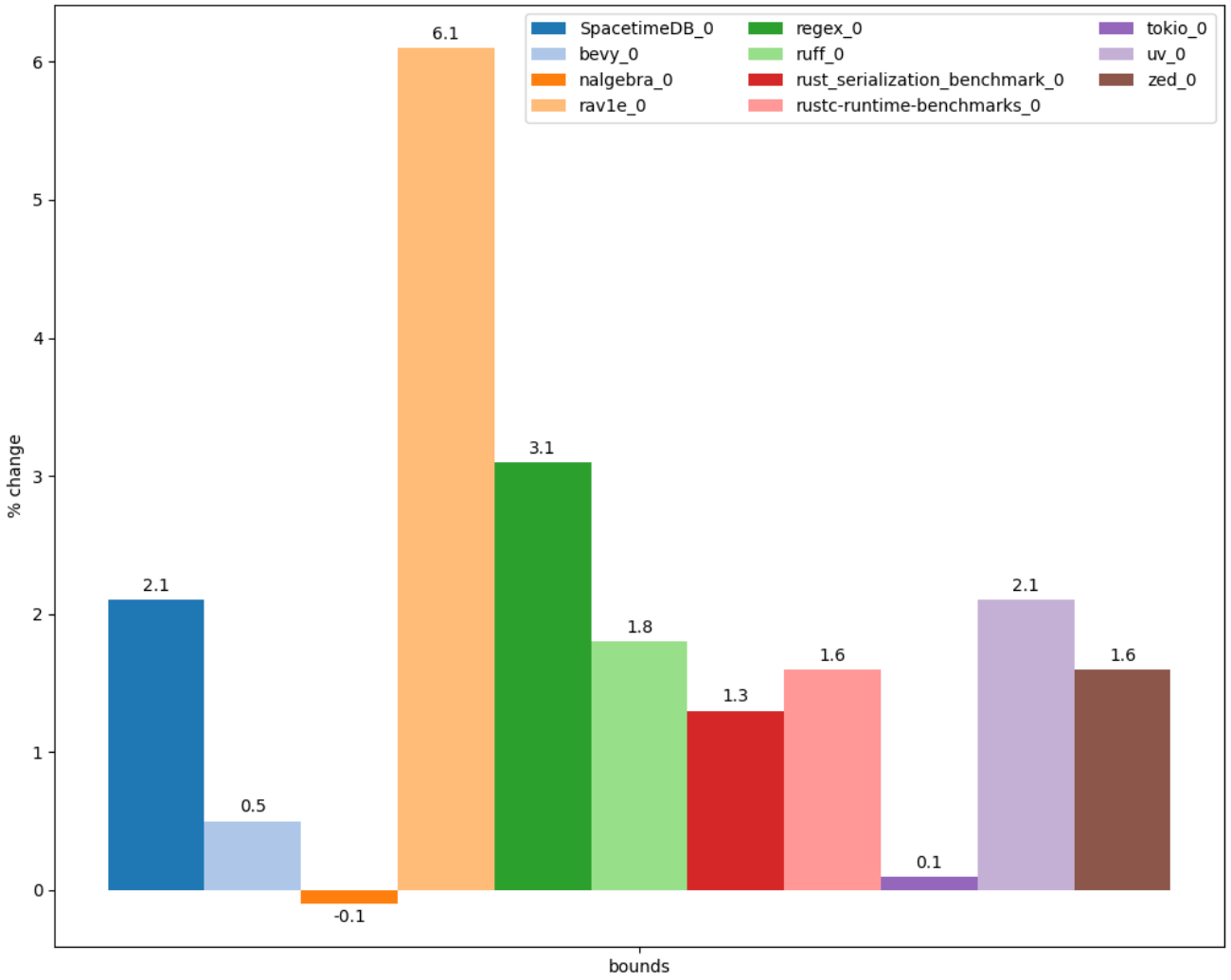
Prevalence

- 20% of checks in Rust compiler are bounds checks
  - Based on analysis of binary code
- In inner loops checks 80% contain only bounds checks
  - These checks prevent vectorization of surrounding loops
  - Data collected with LLVM plugin based on rustc and oxipng analysis

Analysis methodology

Compiler branch

Performance improvement after disabling bounds checks

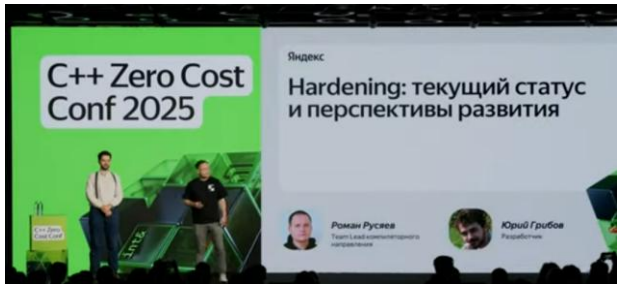


We disabled instrumentation in user code and checks in stdlib (containers, slices, strings, etc.)

[Compiler branch]

Hardened C/C++

- C/C++ provides limited support for memory safety:
  - StackProtector
  - _FORTIFY_SOURCE
  - Hardened STL (C++ only)
  - -fsanitize=bounds, -fsanitize=object-size
- Complete, Rust-like correctness checks are impossible due to lack of information about lengths for raw pointers
  - E.g. only 20% of non-trivial memory accesses in Clang code can be checked
  - C++ Safe Buffers may solve this but requires non-trivial code mofications
- More information:
  - Hardening: current status and trends (C++ Zero Cost 2026)



Mario Simoes, https://snl.no/Mount_Everest

Rust vs. Hardened C++

```rust
pub fn foo(x: &[i32], i: usize) -> i32 {
    x[i]
}
```

rustc -O -Cpanic=abort

https://godbolt.org/z/jGT6adEYM

```
foo:
  cmp     rdx, rsi
  jae     .LBB0_2
  mov     eax, dword ptr [rdi + 4*rdx]
  ret
.LBB0_2:
  push    rax
  lea     rax, [rip + .Lanon.1]
  ...
  call    qword ptr [rip + panic_bounds_check]
```

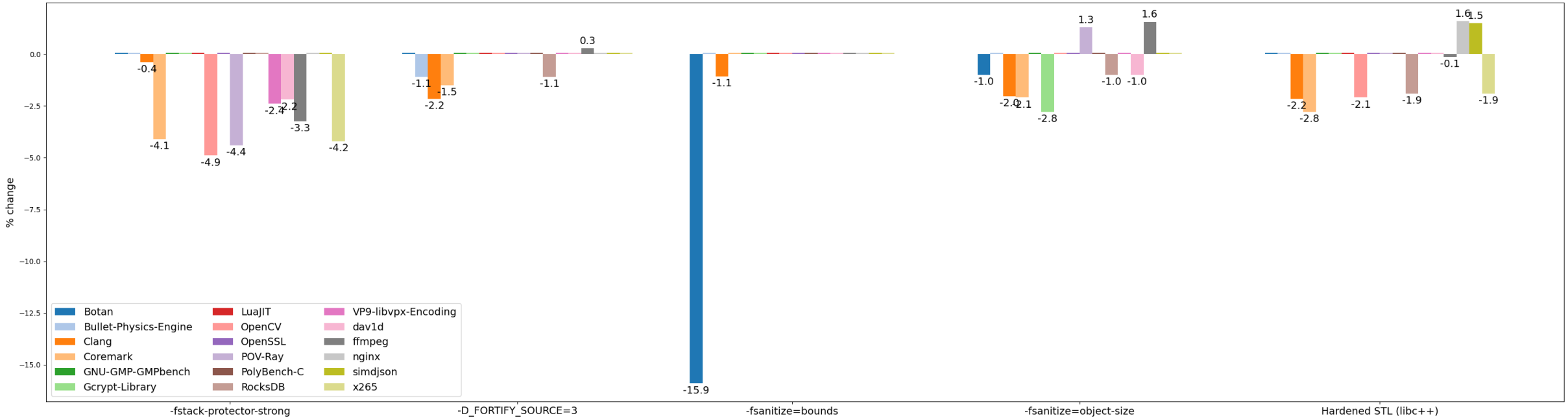
```cpp
extern "C"
int foo(std::span<int> s, size_t i) {
  return s[i];
}
```

gcc -O2 -D_GLIBCXX_ASSERTIONS

https://godbolt.org/z/6MG5vdefK

```
foo:
.LFB821:
  cmp     rdx, rsi
  jnb     .L3
  mov     eax, DWORD PTR [rdi+rdx*4]
  ret
.foo.cold:
.L3:
  push rax
  ...
  call    std::__glibcxx_assert_fail
```

Overheads in Hardened C++



Unexpected performance improvements due to optimizer anomalies (bad vectorization, enabled inlining, etc.)

Conclusions

- Bounds checks in Rust are applied much more aggressively than in Hardened C/C++
  - Hardened C/C++ checks 5x accesses than Rust
- Overheads may reach 3-4% (and are comparable to Hardened C/C++)
  - But some tests may degrade by 10% and more (same applies to Hardened C/C++)
- There are many ways to avoid checks in hot code
  - Some of them won't be needed if delayed panics are finally implemented (rust #50759)

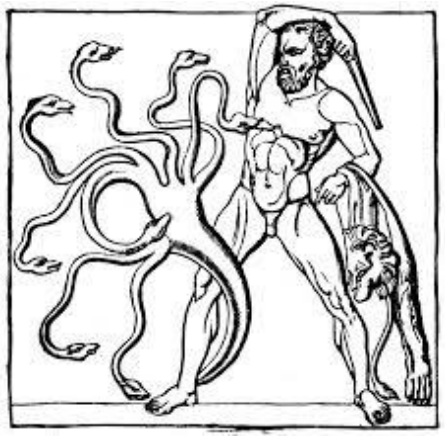
Recommended readings

- How to avoid bounds checks in Rust without unsafe (Sergey Davidoff aka Shnatsel, Rust Secure WG lead)
  - “Bible” of bounds check elimination
- Story-time: C++, bounds checking, performance, and compilers (Chandler Carruth, SW foundations & PLs technical lead, creator of Carbon language)

Aliasing rules

Detailed analysis

Pointer aliasing



- Universal feature in programming languages
- Can two different references (pointers) address same object in program?
- Different languages make different choices:
  - No two pointers in program may refer to same data (e.g. Fortran 77)
  - Any two pointers may address same data (e.g. pre-C89)
  - Intermediate variants
    - E.g. strict (type-based) aliasing in C89 and restrict annotations in C99

https://commons.wikimedia.org/wiki/File:Hercules_slaying_the_Hydra.gif

Why aliasing is important?

- Unlimited aliasing is convenient for developer but reduces optimization opportunities for compiler
- Reduced aliasing enables many optimizations
  - E.g. vectorizers, CSE, GVN, LICM, DSE, etc.
  - Even used in CPU (for speculative execution of loads)
- Alias Analysis is one of the main components of any mature compiler
  - Identifies pointers that may (or may not) alias

```
int alias(int *a, int *b) {
  *a = 0;
  *b = 1;
  return *a;
}

int noalias(int * restrict a, int * restrict b) {
  *a = 0;
  *b = 1;
  return *a;
}
```

https://godbolt.org/z/nqdax4vEP

```
alias:
  mov     DWORD PTR [rdi], 0
  mov     DWORD PTR [rsi], 1
  mov     eax, DWORD PTR [rdi]
  ret

noalias:
  mov     DWORD PTR [rdi], 0
  xor     eax, eax
  mov     DWORD PTR [rsi], 1
  ret
```

Alias analysis theory

- Precise *intra*procedural alias analysis is
  - NP-complete without dynamic allocations (Pointer-induced aliasing, 1991)
  - Undecidable otherwise (Undecidability of static analysis, 1992)
- Aliasing is very rare in real programs
  - 98% of pointers in SPEC benchmarks address only one object
    - Dynamic Points-To Sets (Mock et al., 2001)
- Existing alias analyses are conservative and overestimate aliasing 1.5-20x
  - Dynamic Points-To Sets (Mock et al., 2001)
  - Speculative Alias Analysis for Executable Code (Fernandez et al., 2002)
  - ORAQL — Optimistic Responses to Alias Queries in LLVM (Doerfert et al., 2023)

Aliasing in Rust

- Limitations imposes by Borrow checker simplify alias analysis:
  - Mutable reference may not alias any other (mutable or shared) reference
  - While it exists, mutable reference is the only pointer to object
  - Object addressed by shared reference may not be changed while reference exists
- In practice this achieves similar effect as restrict in C/C++

```
pub fn foo(a: &mut i32,
           b: &mut i32) -> i32 {
    *a = 0;
    *b = 1;
    *a
}
```

```
define i32 @foo(
  ptr noalias ... %a,
  ptr noalias ... %b) {
    store i32 0, ptr %a
    store i32 1, ptr %b
    ret i32 0
}
```

```
foo:
    mov     dword ptr [rdi], 0
    mov     dword ptr [rsi], 1
    xor     eax, eax
    ret
```

https://godbolt.org/z/6WoTzoo5v

Limitations

- Information about aliasing is represented via noalias attribute in function arguments
  - References that appear *inside* functions are not marked and can alias from LLVM standpoint
- May be solved via alias.scope metadata in future (#16515)
- Workaround: create dummy-functions specifically for noalias annotations
  - Example: rustc

```
pub fn foo(a: *mut i32,
           b: *mut i32) -> i32
{
    let a = unsafe { &mut *a };
    let b = unsafe { &mut *b };
    *a = 1;
    *b = 2;
    *a
}
```

```
foo:
    mov     dword ptr [rdi], 1
    mov     dword ptr [rsi], 2
    mov     eax, dword ptr [rdi]
    ret
```

https://godbolt.org/z/WjY9rxex7

Limitations

- One case of above problem is aliasing of references to structure fields
- E.g. when passing two vectors to function compiler fails to understand that their buffers do not alias
- Workaround: pass raw slices to functions

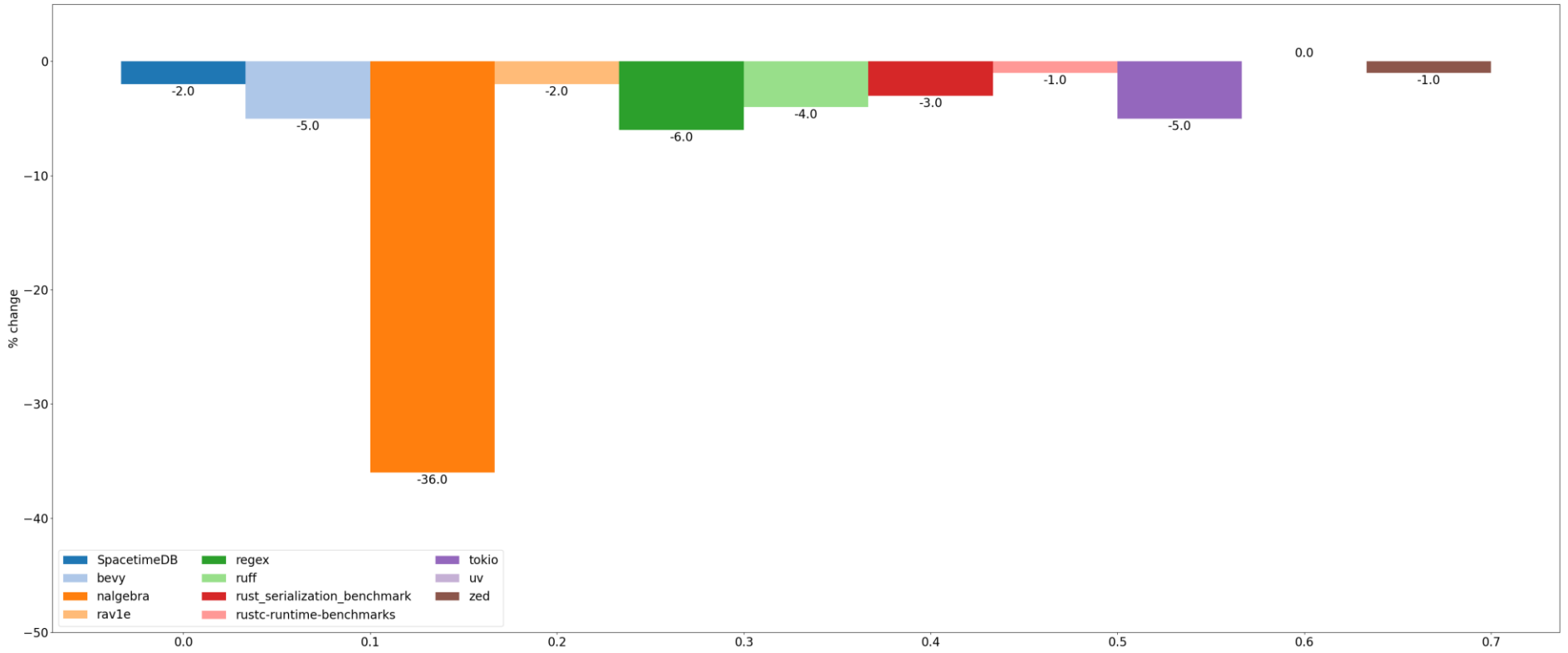
```
pub fn foo(a: &mut Vec<i32>,
           b: &Vec<i32>) {
    let n = a.len();
    let b = &b[..n];
    for i in 0..n {
        a[i] = b[i] + 100;
    }
}
```

```
pub fn foo(a: &mut [i32],
           b: &[i32]) {
    let n = a.len();
    let b = &b[..n];
    for i in 0..n {
        a[i] = b[i] + 100;
    }
}
```

Limitations

- Aliasing rules apply only to references
- Raw pointers will still alias in Rust
  - This is worse than in C/C++ due to lack of strict aliasing rules

Efficiency of Rust aliasing rules



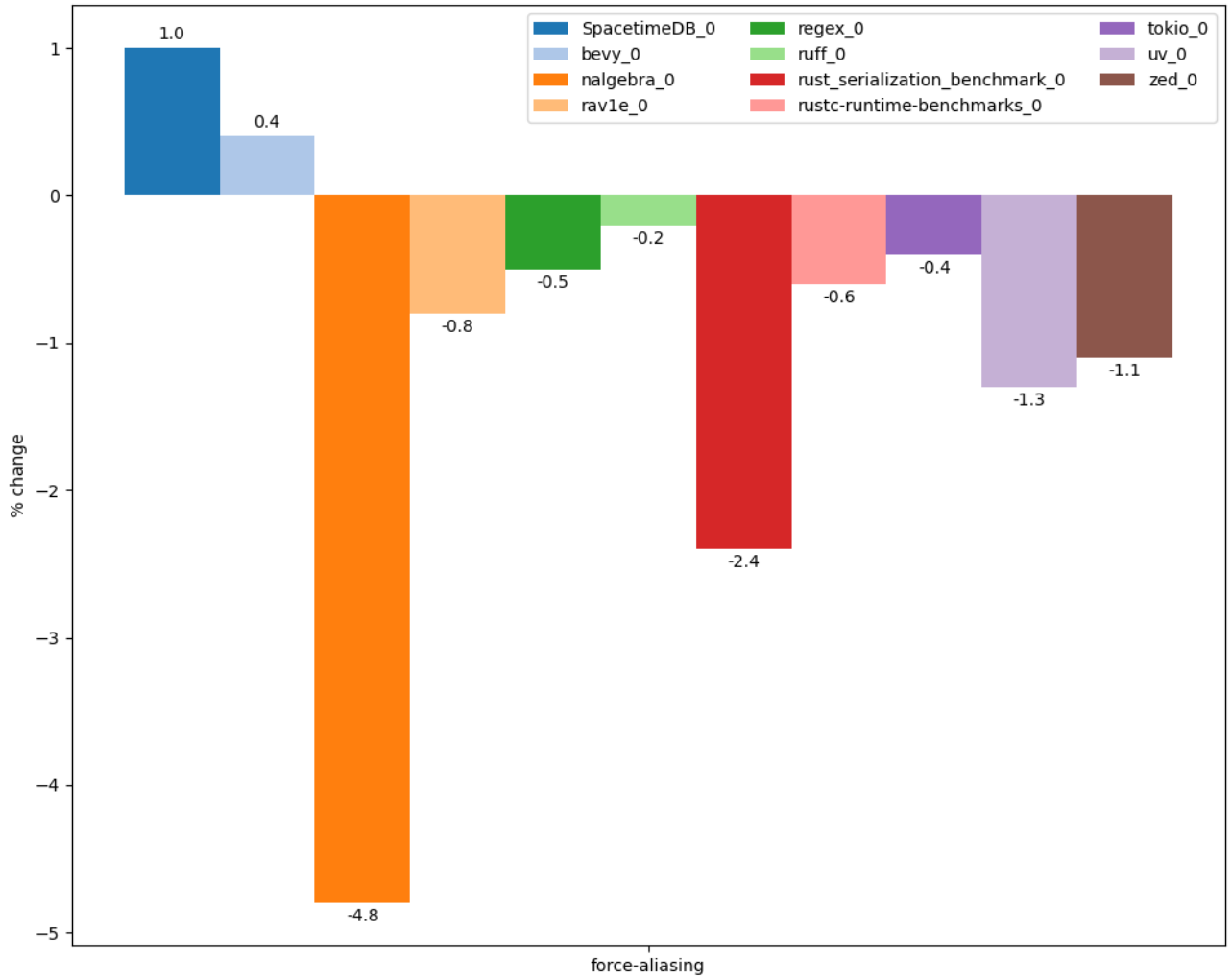
Precision of alias analysis – number of pointer pairs for which analysis could give definite answer (MustAlias / NoAlias)

Degradation of alias analysis precision in Rust after turning of reference rules

[Analysis methodology]

[Compiler branch]

Performance degradation after disabling aliasing rules



[Compiler branch]

Situation in C/C++

- C and C++ disallow aliasing of pointers to different types
  - Except char *, signed/unsigned, scalar/vector, etc.
- But many projects ignore this rule via -fno-strict-aliasing flag

- C and C++ also allow developer to explicitly prohibit aliasing via restrict keyword
  - In practice this is seldomly used (Rust couldn't enable its aliasing rules for long time due to lots of hidden bugs in LLVM)

Conclusions

- Rust aliasing rules allow to
  - Increase analysis precision by 5% on average
  - Improve performance by 1-2% on average
- Can expect performance increase after further codegen improvements (aliasing metadata)

Recommended readings

- Complexity of alias analysis:
  - Pointer-induced aliasing (Landi, 1991)
  - Undecidability of static analysis (Landi, 1992)
- Aliasing in real code:
  - Dynamic Points-To Sets (Mock et al., 2001)
  - Speculative Alias Analysis for Executable Code (Fernandez et al., 2002)
  - ORAQL — Optimistic Responses to Alias Queries in LLVM (Doerfert et al., 2023)
- De-aliasing in CPU:
  - Memory Disambiguation on Skylake

Mandatory initialization

Detailed analysis

Mandatory initialization

- Uninitialized variables may cause information leaks and other errors
  - Password or address leaks (the latter compromises ASLR)
- Prevalence:
  - 10% CVE root cause in Microsoft products in 2018 (see Killing Uninitialized Memory)
  - 12% exploitable bugs in Android (see P2723)

Alex Dos Santos, https://www.pexels.com/photo/vintage-water-tap-dripping-in-black-and-white-29725026/

Example

```
void foo() {
  char password[32];
  ...
}
void bar() {
  char message[1024];
  if (cond) strcpy(message, “...”);
  // Leaking password under !cond
  printf(message);
}
void baz() {
  foo();
  bar();
}
```

Rust approach

- Rust requires initialization of all variables
  - Local, global, heap
- Each variable must be assigned on all paths before first use

```
pub fn foo() {
  let x = 10;
  x
}
```

OK

```
pub fn foo(cond: bool) -> i32 {
  let x;  // No mut needed
  if cond {
    x = 10;
  } else {
    x = 20;
  }
  x
}
```

OK

```
pub fn foo(cond: u32) -> i32 {
  let x;  // No mut needed
  if cond > 10 {
    x = 10;
  } else if cond <= 10 {
    x = 20;
  }
  x
}
```

NOT OK

Other languages

- Many other languages also require initialization (or autoinitialize with zeroes):
  - C#, Swift, Go, Java
  - Hardened C/C++ (-ftrivial-auto-var-init), C++26 (P2795)
    - Only local varibles
      - Global variables and sbrk/mmap-buffers already zero-initialized
      - Heap can be zero-initialized via hardened allocator
    - Enabled by default in Android user/kernel-space and Chrome

Issue with mandatory initializations

- Initializations are often redundant in practice
  - Will be overwritten with other values later without being read
- Compiler can’t always eliminate such redundant initializations
- Example:

```
let mut file = File::open("example.bin")?;

while (...) {
    let mut buffer = [0u8; 1024];  // memset not removed

    let n = file.read_exact(&mut buffer)?;

    ...
}
```

Prevalence of redundant initializations

- How many initializations are redundant?
- Static estimate (via LLVM plugin):
  - Hardened Clang: +4% stores in loops (Clang benchmark)
- Dynamic estimate (via Valgrind plugin):
  - Compiling large C++ source file:
    - Clang: 11% unused writes
    - Hardened Clang (-ftrivial-auto-var-init): 23% unused writes (i.e. 2x increase)

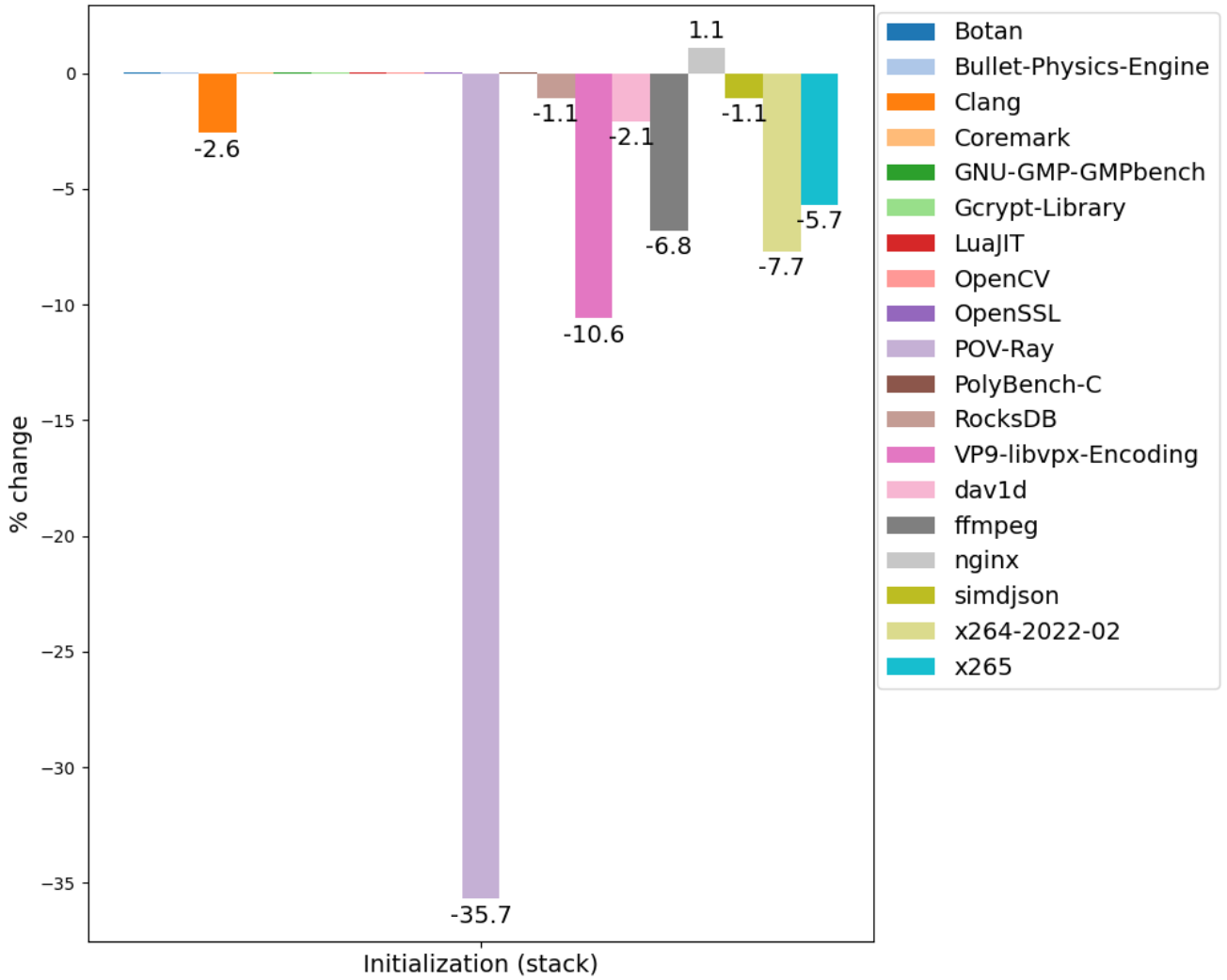
Overheads in benchmarks

- Unfortunately it’s not possible to directly measure initialization overheads in Rust benchmarks
  - That would require extensive code rewrites
- We will illustrate overheads
  - Reported in various Rust projects
  - Measured for Hardened C/C++

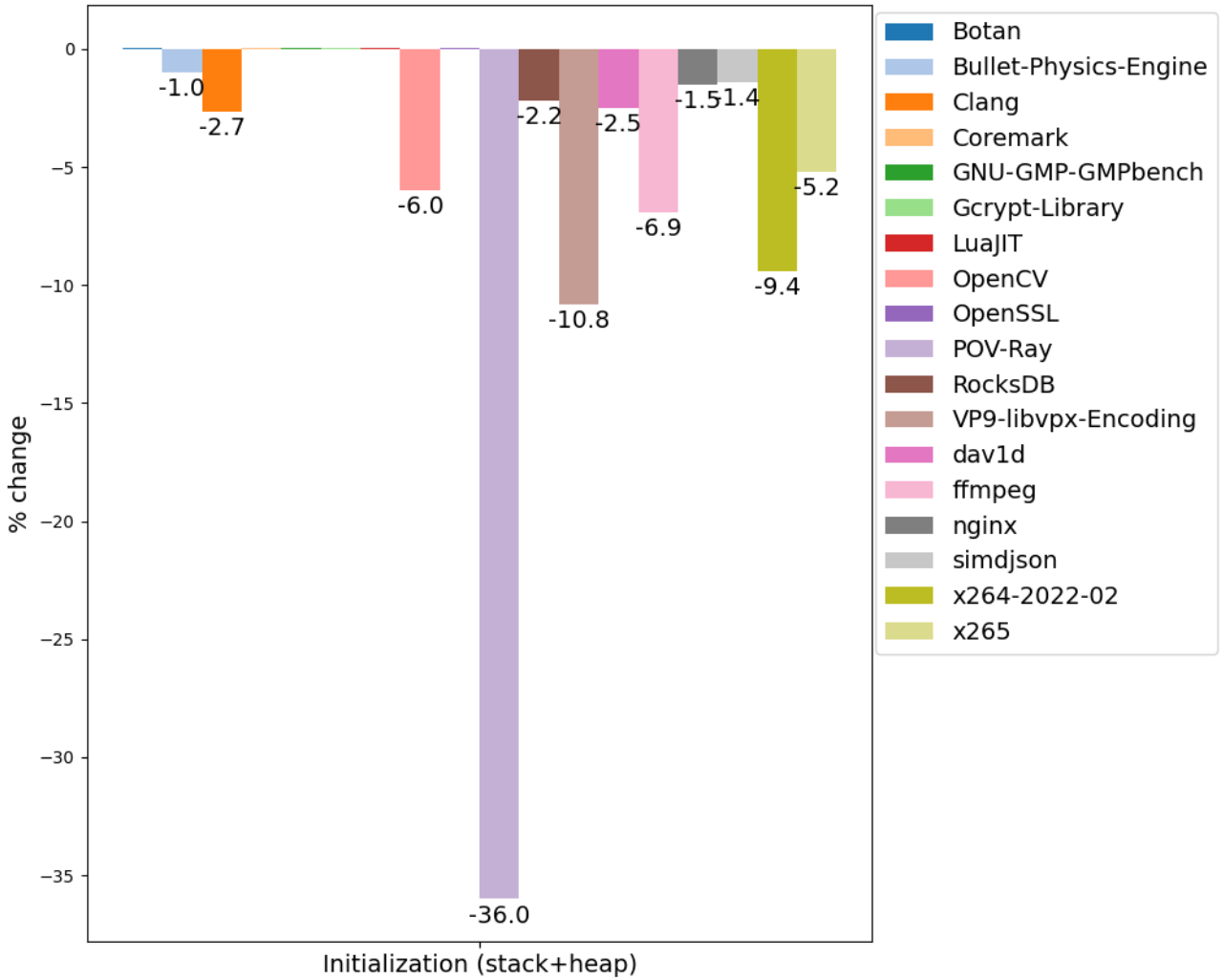
Example overheads in Rust programs

- What are the potential consequences for performance?
  - 7% in stdlib file IO (rust #26950)
  - 20-30% in Hyper HTTP (see tokio #1744)
  - 30% in Shadow simulator (shadow #1643)
  - 25% in stdlib TcpStream (RFC #837)
  - 1.5% in rav1d (Making the rav1d Video Decoder 1% Faster)

Overheads with forced stack initialization for C++



Overheads with forced stack+heap initialization for C++



- We used crude approach to heap initialization (LD_PRELOAD + memset)
- More sophisticated implementation would have smaller overheads

Compiler optimizations

- Redundant scalar initializations can be optimized away by many SSA passes in LLVM (and are very cheap anyway)
  - DCE, BDCE, ADCE, InstCombine, etc.
- Redundant array initializations pose the main problem
  - DeadStoreElimination, MoveAutoInit, MemCpyOptimizer
  - To a lesser degree InstCombine

Workarounds

- One does not simply create a reference to uninitialized object and fill it:

  “Creating a reference with &/&mut is **only allowed if** the pointer is properly aligned and **points to initialized data**” (addr_of_mut)

- Recommended solution is to use MaybeUninit type and various API around it
- Unfortunately MaybeUninit APIs are not always convenient

Workarounds: Standard types

Arrays:

```
let buf :: [u8; 4] = unsafe {
    let mut buf = MaybeUninit::<[u8;
4]>::uninit();
    let ptr = buf.as_mut_ptr();
    for i in ... {
        ptr.offset(i).write(...);
    }
    buf.assume_init()
};
```

Structures:

```
let role :: Role = unsafe {
    let mut uninit =
MaybeUninit::<Role>::uninit();
    let ptr = uninit.as_mut_ptr();
    // Need write
    // (instead of explicit assign)
    // to avoid dropping
    // uninitialized String
    addr_of_mut!((*ptr).name)
        .write("basic".to_string());
    (*ptr).flag = 1;
    (*ptr).disabled = false;
    uninit.assume_init()
};
```

Workarounds: Vectors

```
let mut buf = Vec<u8>::new();

let old_len = buf.len();
buf.reserve(data.len());

unsafe {
    let ptr = buf.spare_capacity_mut();
    for i in ... {
        ptr[idx].write(...);
    }
    buf.set_len(len + data.len());
}
```

Workarounds: IO

```
let mut buf = [MaybeUninit::uninit(); 4096];
let mut buf = BorrowedBuf::from(&mut buf[..]);

rd.read_buf(buf.unfilled())?;

let data : &[u8] = buf.filled();

// Work with buf.filled()
```

Conclusions

- Initialization overheads in Rust and Hardened C++ are on par
  - Higher than in C++26 due to heap initializations
- These overheads can be large in some scenarios (5-10% or more) and require manual code tuning:
  - Using dedicated APIs like MaybeUninit

Recommended readings

- Killing Uninitialized Memory (video)
  - Report from Microsoft about enabling similar checks in Visual Studio
- P2795: Erroneous behavior for uninitialized reads
  - To be integrated to C++26

Symbol localization

Detailed analysis

Symbol localization

- Functions in translation unit (crate in Rust, cpp file in C++) can be used locally or globally (in other translation units)
  - Controlled with static keyword and anonymous namespaces in C++
  - Or with pub, pub(crate), etc. in Rust
- Rust and C++ make opposing default choice:
  - In Rust functions are local for their module by default (top-level functions are local to their crate)
  - pub(crate)-functions are local to crate
  - Only pub-functions are visible across whole program

Localization in Rust

```rust
pub(crate) fn foo1(x: i32) {
    println!("Hello world {}", x);
}

mod mymod {
    pub(super) fn foo2(x: i32) {
        println!("Goodbye world {}", x);
    }

    pub fn foo3(x: i32) {
        println!("Nice world {}", x);
    }
}

pub use mymod::foo3;

pub fn bar() {
    foo1(1);
    mymod::foo2(2);
    mymod::foo3(3);
}
```

```
$ rustc +baseline test.rs -O --emit=asm --
crate-type=rlib -o- \
    | rustfilt \
    | grep globl
        .globl  test::mymod::foo3
        .globl  test::bar
```

Only foo3 and bar
are exported!

Compiler optimizations

- Local functions are much better optimized by compiler
- Usually such optimizations are related to API or ABI changes:
  - Interprocedural constant propagation (IPSCCP)
  - Replacing pass-by-reference with pass-by-value (ArgumentPromotion)
  - Removing dead arguments or return values (DeadArgumentElimination)
  - Relaxed calling convention (using coldcc in Transforms/IPO/GlobalOpt.cpp)
  - Interprocedural register allocation (IPRA)
- And also
  - More aggressive inlining (e.g. local functions which are only called once are always inlined)
  - Better dead code elimination

Linker optimizations

- For shared libraries (DLLs) local functions
  - May be called directly (bypassing PLT/GOT)
  - Do not need to be resolved by loader at startup
- See more details in Dynamic libraries and how to optimize them

Example

```rust
#[inline(never)]  // Simulate inliner fail
fn foo(b: bool, seed: i32, s: &[i32]) -> i32 {
    let mut ans = seed;
    for &x in s {
        ans += if b { 1 } else { ans ^ x }
    }
    ans
}

pub fn bar(n: i32, s: &[i32]) -> i32 {
    let mut ans = 0;
    for i in 0..n {
        ans += foo(true, i, s);
    }
    ans
}
```

https://godbolt.org/z/MM7Tchx49

```
foo:
        lea     eax, [rsi + rdi]
        ret
```

Loop in foo was optimized out

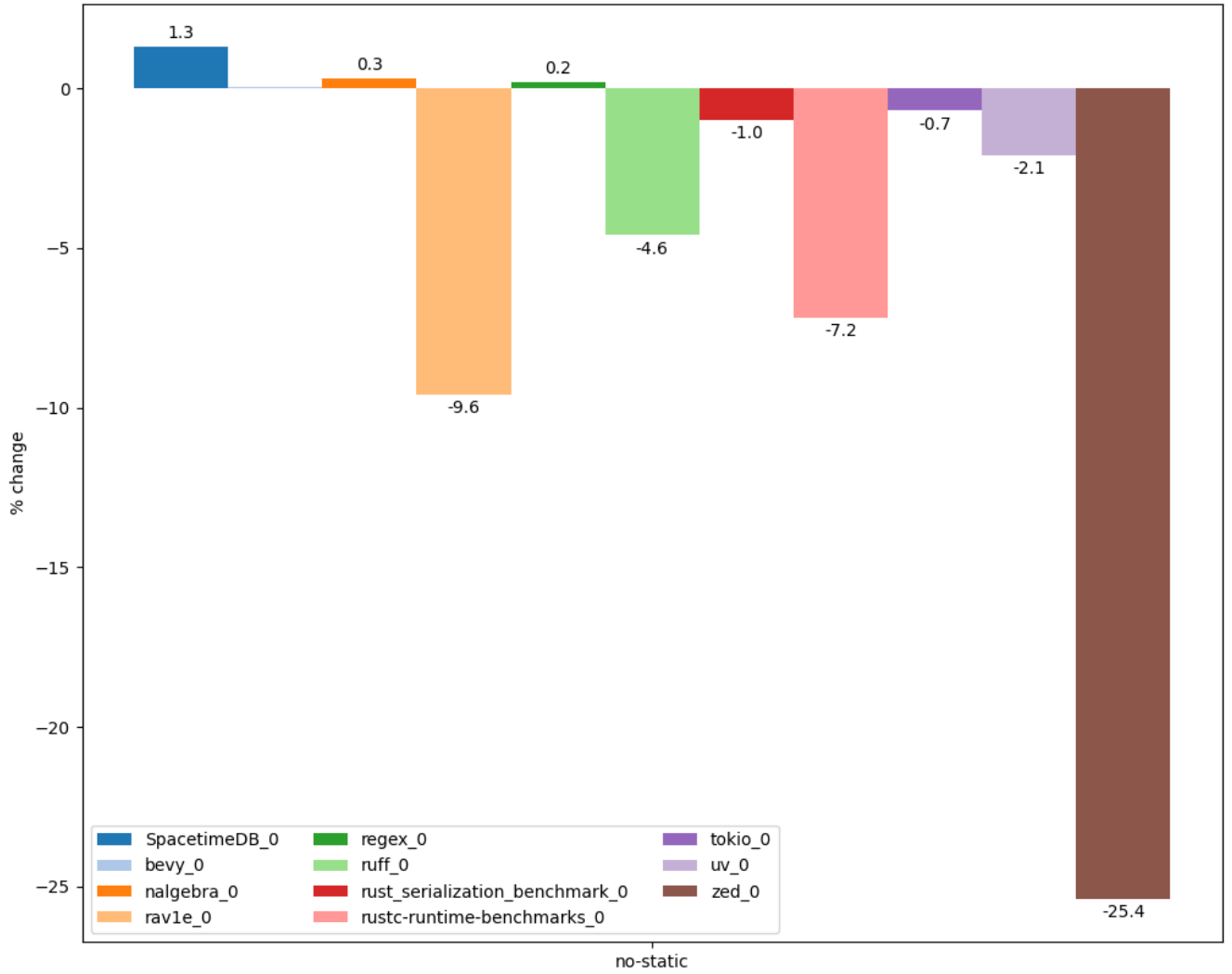
Comparison to C/C++

- Many authoritative coding guidelines suggest to localize functions:
  - Google C++ Style Guide
  - LLVM Coding Standard (Restrict Visibility)
  - C++ Core Guidelines
  - Chromium C++ Style Guide
- No way to localize private class methods in C++
  - Prototype optimization

Prevalence

- 80-90% of functions in Rust programs are local
  - Statistics collected via this patch
- In C/C++ results depend on the project:
  - 80-90% (clang, ffmpeg, git, opencv, bitcoin)
  - 40-50% (openssl, tmux, gcc, libuv)
  - (Statistics collected with Localizer tool)

Slowdown with disabled localization



[Compiler patch]

Conclusions

- Function localization allows compiler to generate much more performant code (over 5%)
- Rust has more optimal performance-wise defaults

Recommended readings

- Efficient C Tips #5 – Make ‘local’ functions ‘static’

No inplace-initialization

Detailed analysis

Overview

- C++ has placement-new mechanism for constructing objects in arbitrary memory buffer:

```
void *addr = some_alloc(sizeof(A));
A *ptr = new(addr) A(...);  // Or std::construct_at
```

- Complemented by perfect forwarding:

```
std::vector<LargeStruct> vec;
vec.emplace_back(large_struct_initializer);
```

- Rust lacks similar mechanism
  - Caused by mandatory object initialization
- Problem:

```
// Can compiler eliminate copy (from stack to heap)?
let b = Box::new(LargeStruct { ... });
```

Why placement new is needed?

```rust
const SIZE: usize = 1024 * 1024 * 1024;

fn main() {
    let b: Box<[usize]> = if cfg!(fast) {
        vec![15; SIZE].into_boxed_slice()
    } else {
        // Problematic copy
        Box::new([15; SIZE])
    };
    black_box(&b);
}
```

```
# Can compiler eliminate copy?
$ LIMIT=$((8 * 1024))
$ rustc +baseline -O test.rs
$ (ulimit -s $ LIMIT; ./test)
thread 'main' has overflowed its stack
fatal runtime error: stack overflow

# Works with dedicated non-copying API
$ rustc +baseline --cfg fast -O test.rs
$ (ulimit -s $ LIMIT; ./test)
```

Why placement new is needed?

```
struct A {
  std::array<int, 1024> vals = {};

  A() = default;
  A(const A &) = default;

  A(int val) {
    std::fill(vals.begin(), vals.end(), val);
  }
};

int main() {
  std::vector<A> aa;
  aa.reserve(NITER);

  for (size_t i = 0; i < NITER; ++i) {
#ifdef USE_PLACEMENT_NEW
    aa.emplace_back(i);
#else
    A a(i);
    aa.push_back(a);
#endif

    asm("" :: "r"(&aa) : "memory");
  }

  return 0;
}
```

USE_PLACEMENT_NEW
~10% faster!

Compiler optimizations

- Move/copy elision
  - Detects and removes redundant moves and copies
- Can work more aggressively (compared to C++)
  - High-level IR (MIR)
  - Shallow move/copy (essentially move/copy in Rust is just a memcpy)
- Modern C++ compilers do not support complete copy elision optimization
  - Only limited optimizations like RVO, NRVO, etc.
  - Mainly due to lack of high-level IR (ClangIR to the rescue?)
  - See Ultimate Copy Elision work by Roman Rusyaev (GitHub)
- Implementation in compiler:
  - MIR (high-level IR): CopyProp, DeadStoreElimination, DestinationPropagation, RenameReturnPlace
  - LLVM: MemCpyOptPass

Detailed analysis

Issues with move/copy elision

- Depends on compiler version
  - Not always monotonically
- Works only for release builds
  - Copies in debug builds are not optimized and may cause stack overflow or significant slowdown
- Language support for inplace-initialization is one of project goals in 2026:
  - In-place initialization

Workarounds

- Same as for mandatory initialization:
  - Use dedicated APIs to create uninitialized buffers and then initialize them:
  - E.g. Box/Rc::new_uninit, Vec::with_capacity/set_len
- Also other APIs e.g. Vec::into_boxed_slice

Conclusions

- Copy elision has more opportunities in Rust than in C++
- But can’t fully compensate for lack of language support for inplace initialization
  - Like placement new and perfect forwarding
- Language developers are aware of this problem and try to solve it:
  - Rust Project Goals 2026: In-place initialization
  - Rust Project Goals 2026: MIR move elimination

Recommended readings

- Ultimate Copy Elision
  - Comprehensive overview of C++ copy elision by Roman Rusyaev

Structure optimizations

Detailed analysis
(not filled yet)

Alignment of structure fields

- Many languages (C, C++, C#, Swift) preserve order of fields in structures according to source code
- Fields need to be properly aligned by adding padding bytes in between
  - Alignment is necessary due to HW constraints
  - Unaligned memory access may cause exception or slowdown
- Alignment speeds up access to fields but reduces D$ locality

```
struct RandomOrder {
  uint8_t  m_byte1;
  uint64_t m_long;
  uint8_t  m_byte2;
  uint32_t m_int;
  uint8_t  m_byte3;
};
```

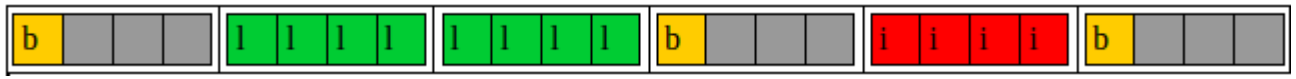


Image source

Structures in Rust

- Rust ABI does not guarantee order of fields in structs
- This allows compiler to freely modify it
- Fields can be reordered to
  - Reduce D$ pressure (by reducing amount of padding)
  - Optimize placement of niches (see below)
- In theory compiler could also co-allocate fields that are frequently used together
  - Could be enabled via PGO or static analysis (like -fipa-struct-reorg in GCC)
- And extract rarely used fields (structure peeling)

Layout optimizations

- Struct fields are sorted according to decreasing alignment to reduce struct size

```rust
pub fn fun(val: Foo) -> i64
{
    val.a
        + (val.b as i64)
        + (val.c as i64)
}
```

```rust
pub struct Foo {
    pub b: bool,
    pub a: i64,
    pub c: i16,
} // 16 bytes total
```

```
fun:
  movzx   ecx, byte ptr [rdi + 10]
  add     rcx, qword ptr [rdi]
  movsx   rax, word ptr [rdi + 8]
  add     rax, rcx
  ret
```

https://godbolt.org/z/MxYh7bv34

```rust
#[repr(C)]
pub struct Foo {
    pub b: bool,
    pub a: i64,
    pub c: i16,
} // 24 bytes total
```

```
fun:
  movzx   ecx, byte ptr [rdi]
  add     rcx, qword ptr [rdi + 8]
  movsx   rax, word ptr [rdi + 16]
  add     rax, rcx
  ret
```

https://godbolt.org/z/GMoejn7rd

Jeffrey Bary, https://www.flickr.com/photos/70118259@N00/3173642531

C++ status

- Many static analyzers detect suboptimal field order:
  - -Wpadded, clang-tidy, PVS-studio, etc.
  - Pahole (uses debuginfo)
- clang-reorder-fields – LLVM-based tool for reordering fields to minimize padding
- In theory with Fat LTO compiler could reorder fields automatically
  - Already done in LCC compiler for Elbrus

Internals of Rust enums

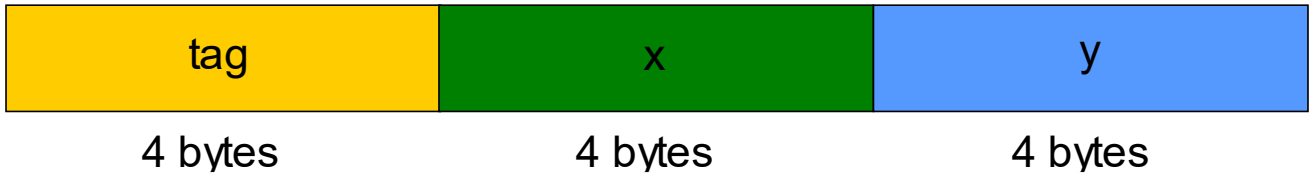
- Rust enums are analogous to std::variant in C++
- The may contain different types, identified by hidden tag field
- Enum object consists of
  - Tag (or discriminant) – index of actual type
  - This type's data

```
pub enum Foo {
    A {x: i32, y: i32},
    B {x: i32, y: i32},
}

pub fn fun(val: Foo) -> i32 {
    match val {
        Foo::A {x, ..} => x,
        Foo::B {y, ..} => y,
    }
}
```

https://godbolt.org/z/n8a1x9nWr

```
fun:
  mov     eax, dword ptr [rdi]
  mov     eax, dword ptr [rdi + 4*rax + 4]
  ret
```



Niches

- Some bit combinations may not correspond to any valid value of encoded type
- For example
  - &x or NonZero<i32> may not be zero
  - bool (8 bits) can only be 0b0 or 0b1
    - Remaining 254 values are invalid
  - char (32 bits) may only contain values in [0x0, 0xD7FF] and [0xE000, 0x10FFFF] ranges
- Such invalid value ranges (not bits!) are called niches
- They may be used to keep tag of enum types (instead of using separate field)
  - So called Discriminant Elision

https://nowmediagroup.tv/blog/10-must-read-books-why-multidisciplinary-reading-is-a-must/

Examples of niches

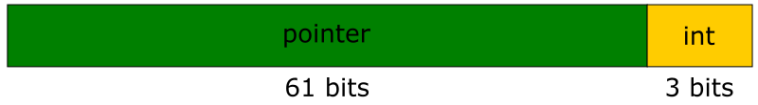
- The most common (and guaranteed by the language) example: Option<T> with T being a NonNull type (e.g. &U)

$$\texttt{Option<\&U>} = \begin{cases} \text{None} & = 0x0 \\ \text{Some(\&u)} & = \text{addr\_of!(u)} \end{cases}$$

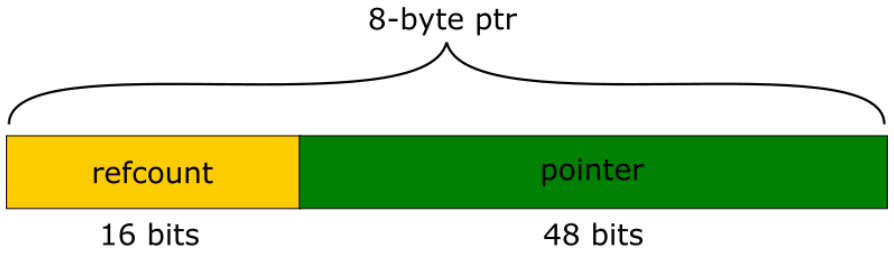
- Same niche may be used for multiple tags:
  - `size_of::<Option<Option<Option<bool>>>>() == 1`
- Similar optimization possible to lower bits of reference (or pointer)
  - Lower bits of aligned pointers are always 0
  - Can be used as niche
  - Currently only supported in nightly (-Zreference-niches, rust #3204)

C++ status

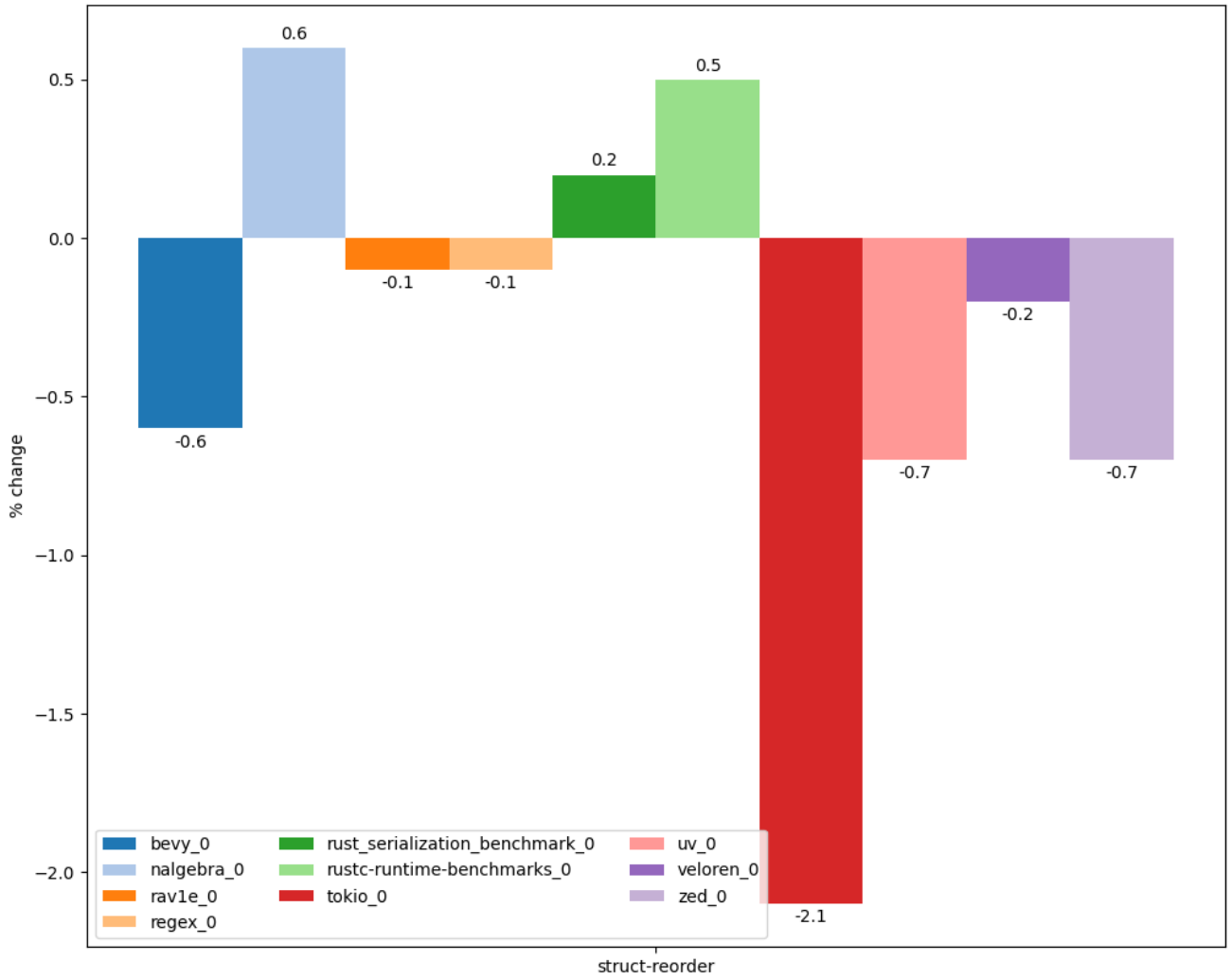
- std::optional and std::variant do not optimize for niches
  - Analog of std::optional with niched tags can be implemented
- llvm::PointerIntPair
  - Uses lower pointer bits to store int value
  - Similar to -Zreference-niches in Rust



- Lock-free implementations of atomic shared_ptr (e.g. in Folly) use free address bits for reference count
  - Allows for shorter and faster atomic operations



Slowdown with disabled struct reorder



- Disabled shouldMergeGEPs and store merging as they were too sensitive to field order

[Compiler patch]

Conclusions

- Rust does not guarantee order of fields in structs
  - Allows more aggressive optimizations
  - And care in unsafe code
- Compiler may aggressively reorder fields and use them for tags
  - Analogous optimizations can be done manually in C++
- Field reordering in itself does not significantly affect performance but can enable or disable other important optimizations
  - Both in compiler and in processor
  - With both positive and negative performance effects

Recommended readings

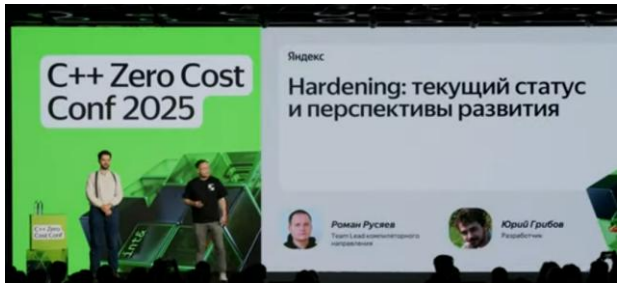
- The Lost Art of Structure Packing (by ESR)

Arithmetic checks

Detailed analysis

Overview

- Arithmetic overflows may cause serious system crashes
  - ~1% CVE and 1.5% KEV in 2024
  - 23 place in Mitre CWE Top 25 2024 (8 in 2019 rating)
- Well-known incidents:
  - Therac-25 radiation therapy machine (1985)
  - Ariane 5 catastrophe (1996)
- History (pre-Rust):
  - `-ftrapv` added in 2000 in GCC (patch for -ftrapv option)
    - Wasn't well maintained and rotted quickly (e.g. BZ #35412 opened 2008 and still not fixed)
  - John Regehr's works on IOC checker in 2010
  - UBsan created in 2014
    - State-of-the-art solution
- Additional info:
  - Hardening: current status and trends (C++ Zero Cost 2026)



J.R. Armstrong, Famous Failures Revisited

Example error (OpenSSH 3.3)

```
nresp = packet_get_int();
if (nresp > 0) {
  // Overflow to zero ...
  response = xmalloc(nresp*sizeof(char*));
  // ... and cause heap buffer overflow here
  for (i = 0; i < nresp; i++)
    response[i] = packet_get_string(NULL);
}
```

Default arithmetic checks in Rust

- Some (not very important...) arithmetic checks enabled by default
  - Division by zero
  - Signed overflow on division (INT_MIN / -1)
  - Float-to-integer saturation
    - Can be disabled via -Zsaturating-float-casts=off
- Integer overflows in stdlib containers
  - Via assert!, checked_add, etc.

Important missing checks

- For historical reasons some important checks are missing
  - And can’t even be enabled via flag
- E.g. overflow check on casing types via as
  - Need to use other APIs like try_into/ try_from

```
fn main() {
    0x100i32 as i8;
}
```

```
$ ./test
0
```

```
use std::convert::TryFrom;

fn main() {
    i8::try_from(0x100i32).unwrap();
}
```

```
$ ./test

thread 'main' panicked at test.rs:5:32:
called `Result::unwrap()` on an `Err` value:
TryFromIntError(())
```

Integer overflow checks in Rust

- By default integer overflows in user code are checked only in debug builds
- Two's complement semantics in release
- Can be enabled in release via -C overflow-checks=on
- Reason: too large overheads
  - "Plan is to turn on checking by default *if* we can get the performance hit low enough and also that it should leave room in the future to move towards universal overflow checking if it becomes feasible" (Niko Matsakis, co-lead of Rust language design team)

Other overheads

- Overheads for overflow checks:
  - Compare + predictable conditional branch (1-2 ALU slots + I$ pressure)
  - Hurt other optimizations (vectorizer, etc.)
- The first (less important) issue could *potentially* be fixed with hardware support

```rust
pub fn foo(xs: &[i32]) -> i32 {
    let mut ans = 0;
    for x in xs {
        ans += x;
    }
    ans
}
```

```
$ rustc ... test.rs
...
.LBB0_5:
        // Loop successfully vectorized
        movdqu  xmm2, xmmword ptr [rdi + 4*rax]
        paddd   xmm0, xmm2
        ...
$ rustc ... -Coverflow-checks=on test.rs
...
.LBB0_3:
        // Loop not vectorized
        add     eax, dword ptr [rdi + rcx]
        jo      .LBB0_6
        add     rcx, 4
        cmp     rsi, rcx
        jne     .LBB0_3
        ...
```

https://godbolt.org/z/1x49jffv3

https://godbolt.org/z/43h1c87M5

Defined overflow

- Unlike C/C++ integer overflows are not Undefined Behavior
  - Panic or Two's complement
- This deprives optimizer of some important information
- Can reduce precision of some analyses (especially ScalarEvolution) and disable some optimizations in LLVM
  - See How undefined signed overflow enables optimizations in GCC

Defined overflow: Example

From Garbage In, Garbage Out (Chandler Carruth)

```
for (;;) {
    int c1 = buf[i1];
    int c2 = buf[i2];
    if (c1 != c2)
        return c1 - c2;
    i1++;
    i2++;
}
```

```
.LBB0_1:
  movzx   eax, byte ptr [rdx + rsi]
  movzx   edi, byte ptr [rdx + rcx]
  inc     rdx
  cmp     al, dil
  je      .LBB0_1
```

https://godbolt.org/z/Yssq68Gja

```
loop {
    let c1 = unsafe {
      *buf.get_unchecked(i1 as usize)
    } as i32;
    let c2 = unsafe {
      *buf.get_unchecked(i2 as usize)
    } as i32;
    if c1 != c2 {
        return c1 - c2;
    }
    i1 += 1;
    i2 += 1;
}
```

Compiler has to assume potential overflow

```
.LBB0_2:
  movsxd  rdi, edi
  movzx   eax, byte ptr [rdx + rdi]
  movsxd  rsi, esi
  movzx   ecx, byte ptr [rdx + rsi]
  inc     edi
  inc     esi
  cmp     al, cl
  je      .LBB0_2
```

https://godbolt.org/z/c1sMcxfsP

Problem with inclusive ranges

- Loops with included upper bound are less optimized by compiler:

```
for i in 1..=n {
    total += i;
}
```

- Due to potential overflow (when `n==T::MAX`) compiler has to perform additional check:

```
while i <= n {
    total += i;
    if i >= n {
        break;
    }
    i += 1;
}
```

- Developers can help compiler by converting to exclusive loop:

```
for i in 1..(n+1)
```

- Or using internal iteration (.for_each, .fold, etc.)

Compiler optimizations

- Integer overflows are not well optimized by compiler
  - E.g. Pr*e*sburger arithmetic is decidable but its complexity is O(2^2^N)
- Only ~30% of arithmetic checks are eliminated when building Rust compiler (with -Coverflow-checks=on)
  - Based on analysis of binary code
- This is the main reason for lack of arithmetic checks in release builds

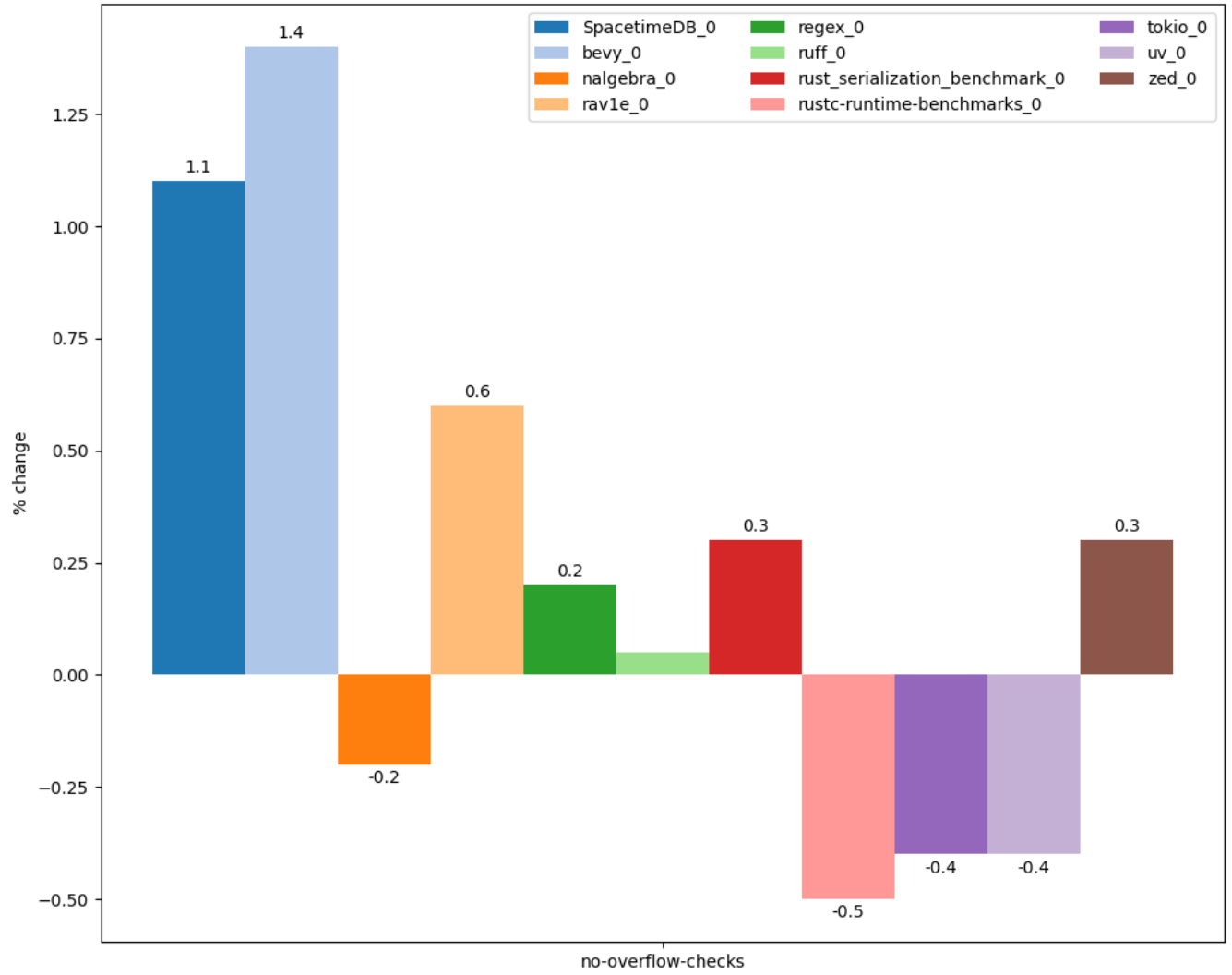
Workarounds

- Unsafe APIs:
  - unchecked_add, wrapping_add, checked_add, etc.
  - to_int_unchecked
  - std::hint::unreachable_unchecked

```
if x <= i32::MIN >> 1 || x >= i32::MAX >> 1 {
    unsafe { std::hint::unreachable_unchecked() }
}
x * 2 / 2
```

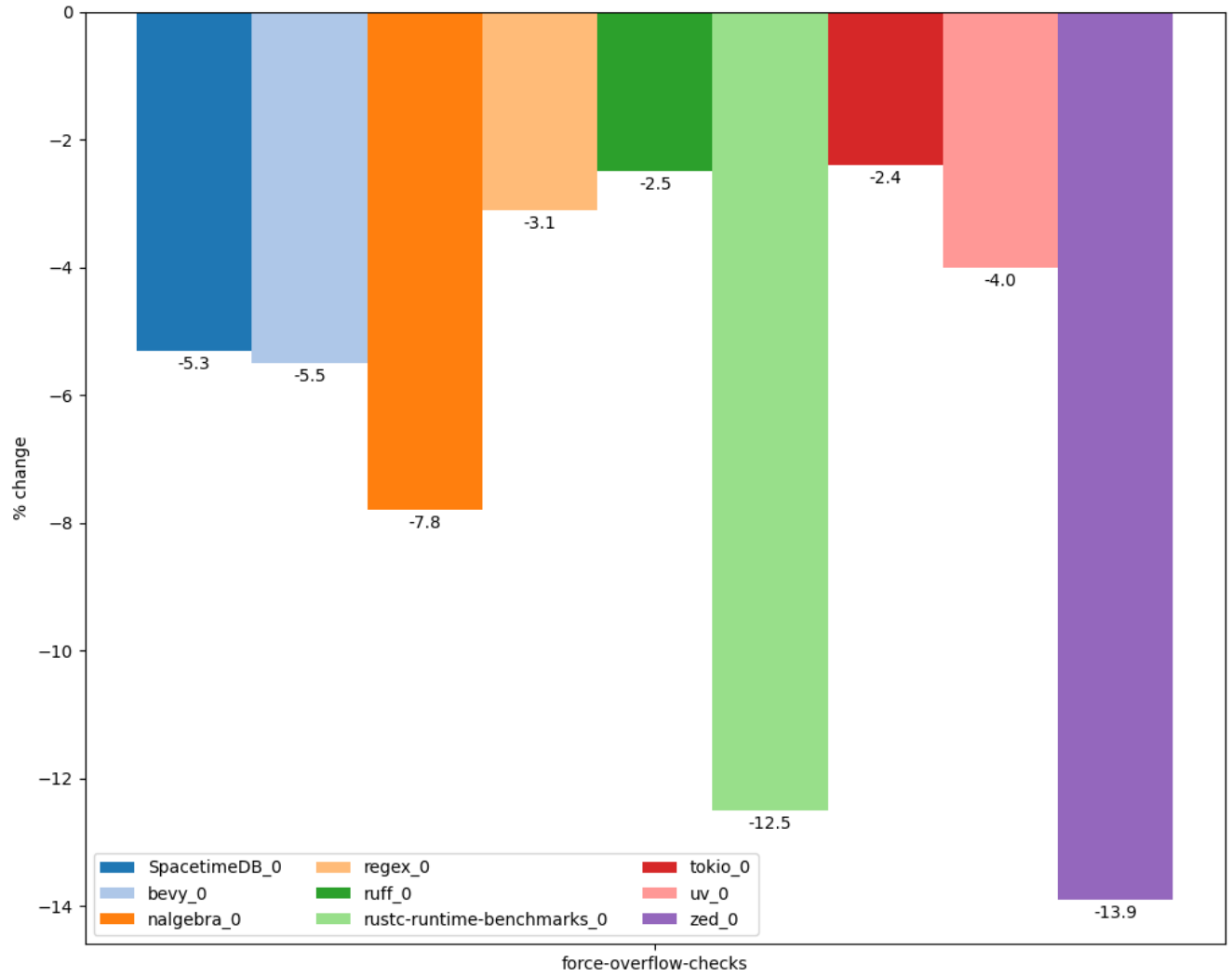
  - (checks in stdlib *can not be disabled*)
- Special stdlib types:
  - NonZeroU32 to disable divide-by-zero checks
  - Wrapping<T> for overflows

Improvements without default checks (division by zero, etc.)



[Compiler branch]

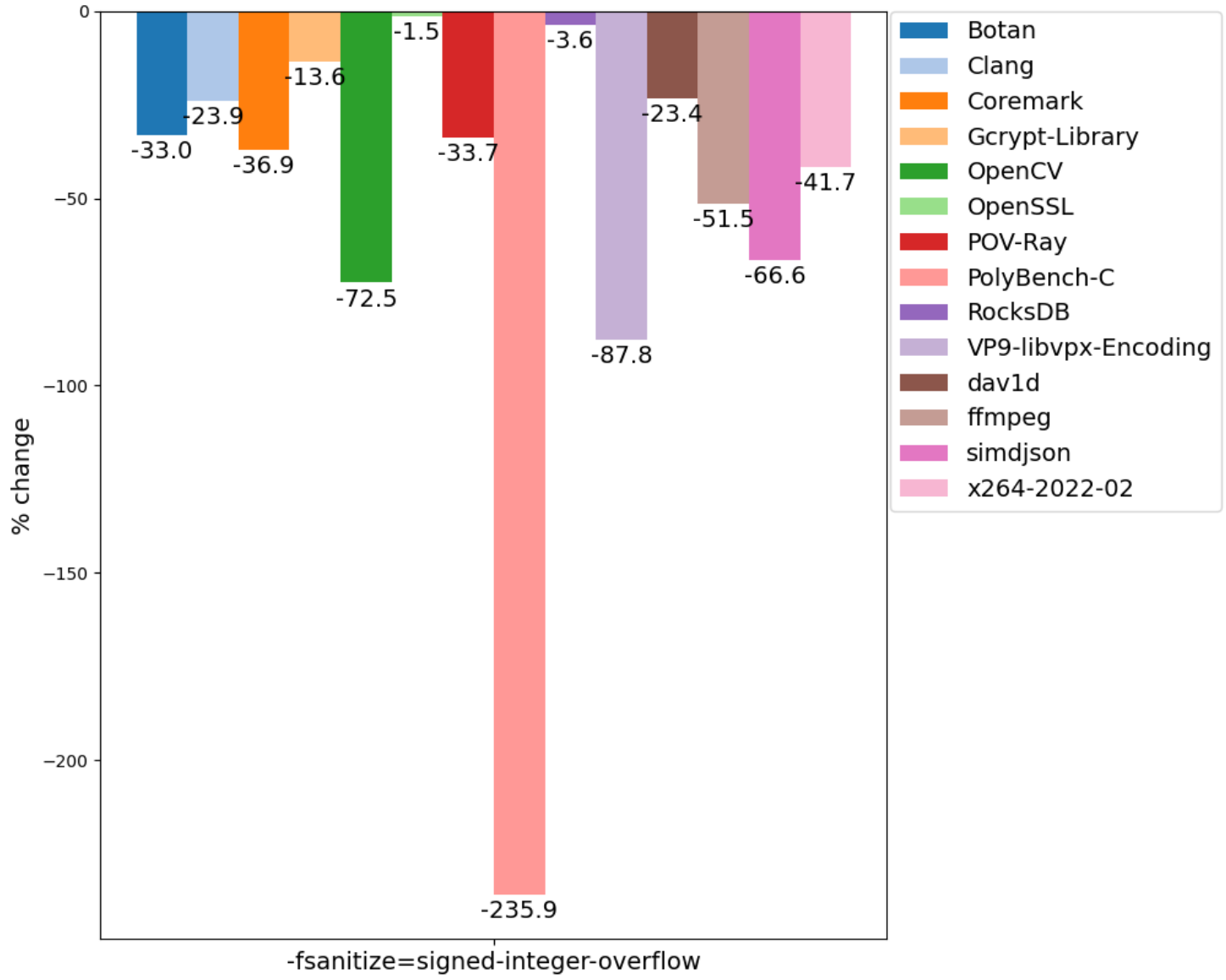
Slowdown with integer overflow checks



- Some tests missing due to crashes
- +30% checks in rustc

[Compiler branch]

Slowdown in C++



- Some tests missing due to crashes
- Even larger overheads in C/C++ with UBsan

Conclusions

- Integer overflow is not UB in Rust
  - Prevents compiler from performing some aggressive optimizations
- Rust makes non-obvious choice about default arithmetic checks
  - Some really important errors like lossy casts are not checked but zero divisions are
  - Rare but may cause slowdown of ~1%
- Integer overflow checks have very large overhead (up to 10%)
  - Same as UBsan
  - Unlikely to be enabled by default in future

Recommended readings

- RFC 560: Integer Overflow (also discussion and overview)
  - Attempt to enable integer overflow checks in release
- Understanding Integer Overflow in C/C++
  - Paper by John Regehr
- How undefined signed overflow enables optimizations in GCC
  - Paper about performance benefits of Integer Overflow UB in C/C++
- Exploiting Undefined Behavior in C/C++ Programs for Optimization: A Study on the Performance Impact
  - Paper about benefits of UB, including signed overflow, with benchmarks

Standard library

Detailed analysis

Overview

- Rust stdlib has more mandatory checks and in some cases uses more robust (and slow) algorithms compared to STL
  - Cargo build system greatly simplifies experiments with alternate implementations though
- On the other hand in other cases it can use faster container and algorithm implementations (compared to STL)

Standard library checks

- Stdlib contains a lot of mandatory checks (assert!, checked_add, etc.)
- Main types of checks:
  - Correctness of indices in containers (bounds checks, already considered above)
  - Size overflow checks in containers (capacity overflow)
  - Successful allocations in containers
  - Correctness of indices in UTF-8 container accesses (&str and String)
  - API-specific checks (comparator axioms, etc.)
- Similar checks are available in STL
  - Under _GLIBCXX_ASSERTIONS and _LIBCPP_HARDENING_MODE (Hardened STL)

Example

```rust
pub fn foo(length: usize) -> Vec<i32> {
    vec![-1; length]
}
```

```
lea     r15, [4*rsi]
xor     r13d, r13d
mov     rax, rsi
// Check that 4*length does not overflow
shr     rax, 62
jne     .LBB0_11
// Check that 4*length <= isize::max
movabs  rax, 9223372036854775804
cmp     r15, rax
ja      .LBB0_11
...
// Check that alloc is successful
call    qword ptr [rip + rustc::rust_alloc]
test    rax, rax
je      .LBB0_11
...
```

https://godbolt.org/z/oa67f1j5z

Example

```rust
use std::io;

fn main() {
    let mut input = String::new();
    io::stdin()
        .read_line(&mut input)
        .unwrap();
    println!("{}", input);
}
```

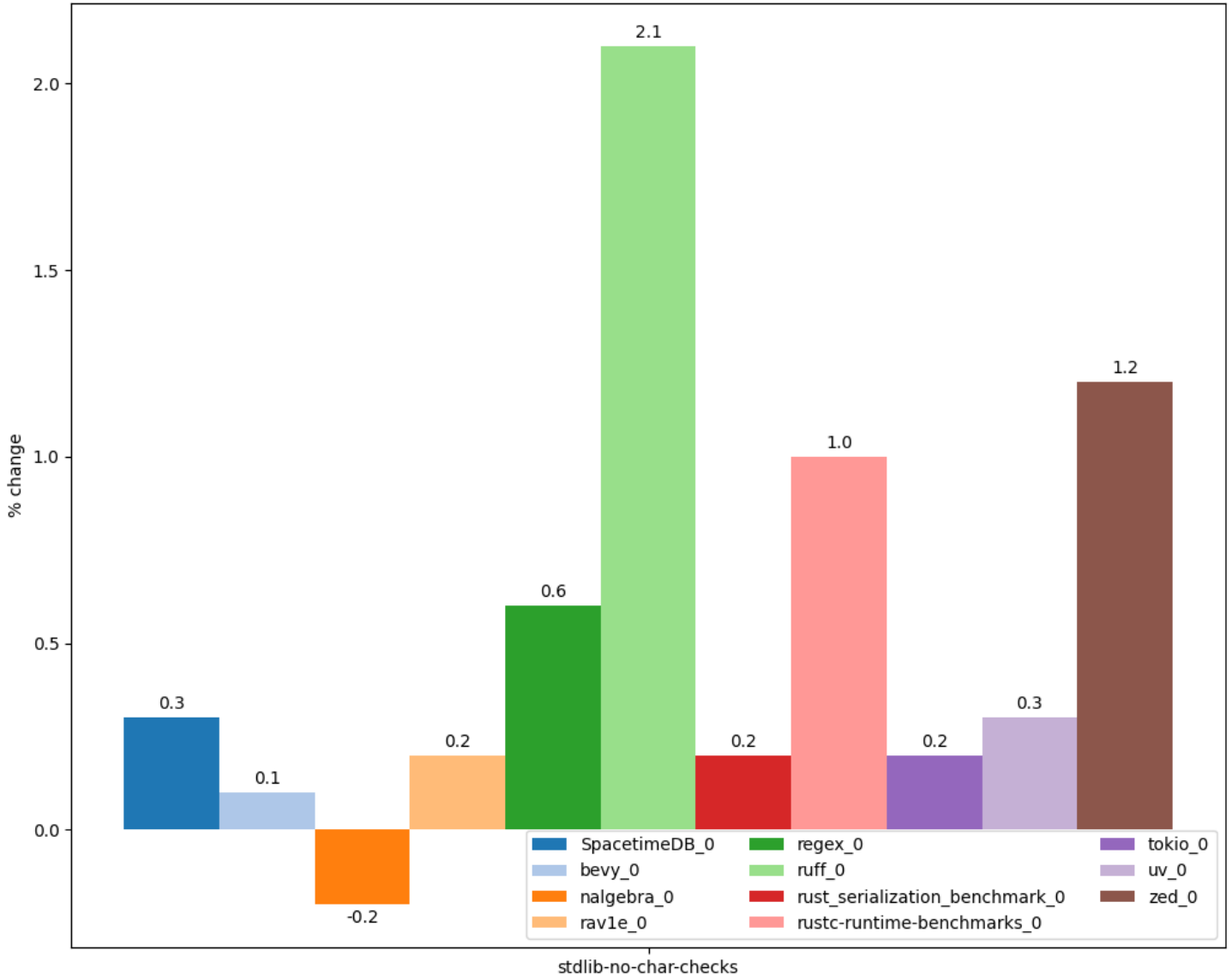
```
$ echo Привет | ./test
Привет

$ echo Привет | iconv -f UTF-8 -t KOI8-RU | ./test
thread 'main' panicked at test.rs:5:39:
called `Result::unwrap()` on an `Err` value:
Error { kind: InvalidData, message: "stream did not contain valid UTF-8" }
```

Other overheads

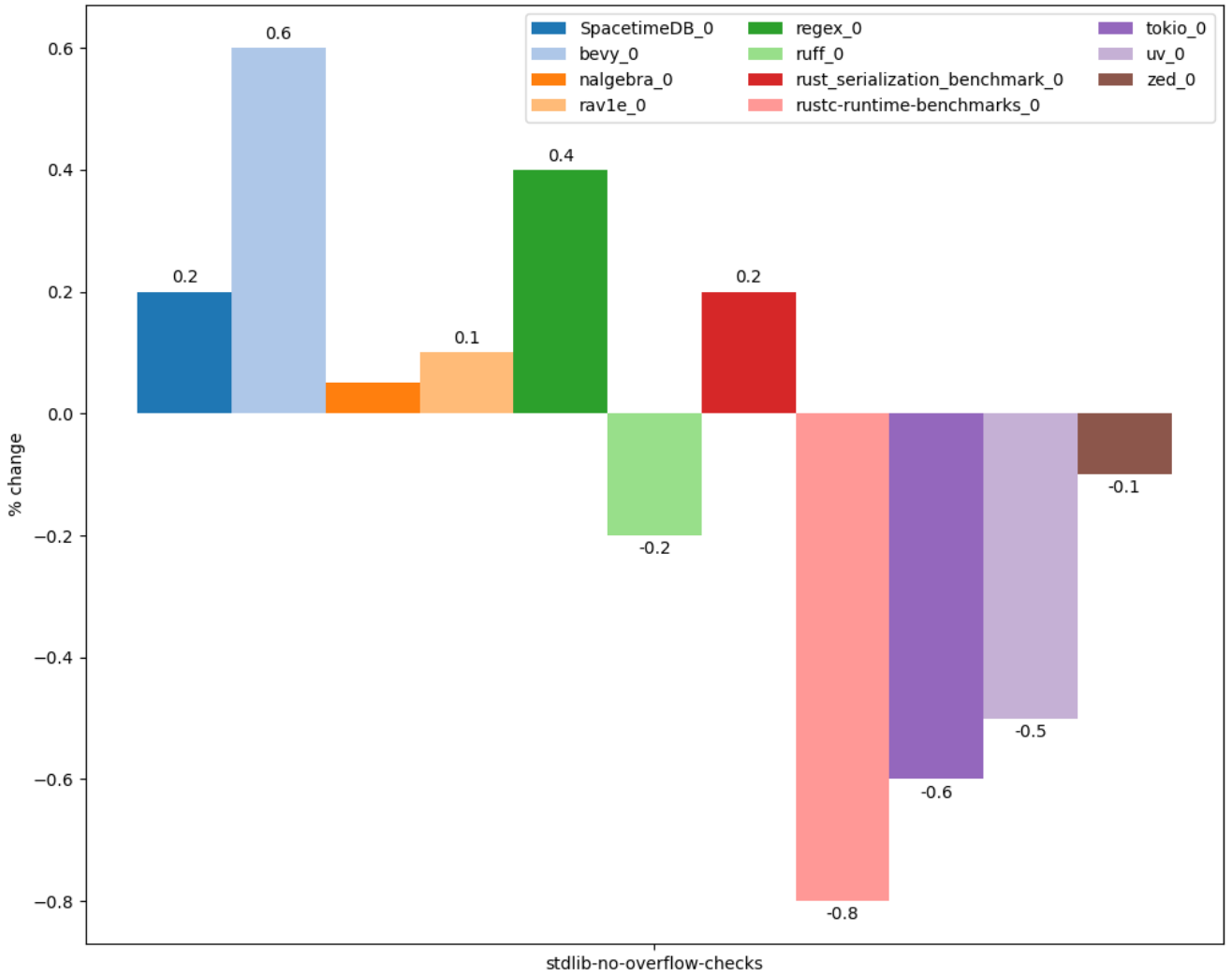
- Default hashing algorithm in hashtables (SipHash) is resilient against DoS attacks bit is slower than analogs in STL
  - 2x slowdown against fxhash in simple microbenchmark
- Rust IO is unbuffered by default
  - Without BufReader programs may be 10x slower
- io::stdout() is line-buffered for non-TTY output (rust #60673)
  - This may significantly slow down command-line tools (for cases like "./prog > file")
- print!/println! macros use locked access to stdout for stable output from parallel threads

Improvements with disabled UTF-8 checks



[Compiler branch]

Improvements with disabled container overflow checks



[Compiler branch]

Optimizations in standard library

- Hashtables:
  - No requirement for element reference validity after modification (as in std::unordered_map)
  - Allows to use open-addressing instead of chaining (fewer allocations, cache-friendly)
- Search trees use B-tree (instead of RB-tree as std::map)
  - Fewer allocations, cache-friendly
  - 4x speedup on simple microbenchmark
- Modern ILP-friendly sorting algorithms (ipnsort, driftsort)
  - Modern STLs are implemented to simplify vectorization when sorting primitive types
  - 65% speedup on simple microbenchmark

Conclusions

- Rust stdlib includes both factors which negatively affect performance and more efficient (over STL) algorithms and containers
- Stdlib issues are not critical due to
  - Availability of many alternative implementations
  - Ease of integration of external libraries in Cargo ecosystem

Recommended readings

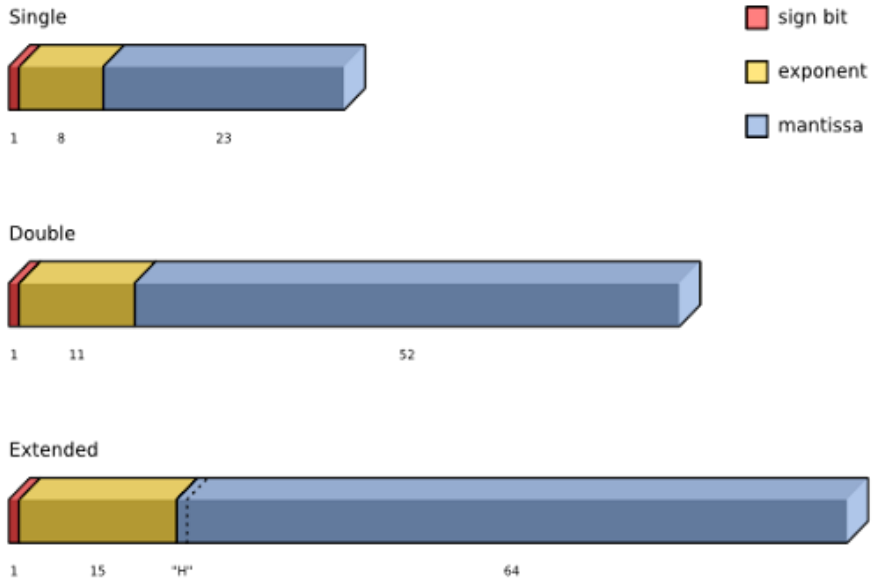
- Safety vs Performance. A case study of C, C++ and Rust sort implementations

Lack of fast-math

Detailed analysis

Floating point computations

- IEEE 754 standard
  - Types and their binary representation
  - Operations on types
  - Rounding
- Most architectures and languages follow the standard (mostly)
- Reasons for discrepancies:
  - Architecture specifics
  - More opportunities for optimizations
- Fast math – umbrella term for optimizations that are enabled under relaxed IEEE 754 rules



Types of fast-math flags

- Algebraic:
  - Associativity
  - Unsignedness of zero
  - Replacing division with multiplication
  - Fusion of operations (e.g. FMA)
- Disallowing non-standard values:
  - NaN, Infinity

```
double foo(double val) {
  return (val + 3.0) / 3.0;
}
```

```
.LCPI0_0:
        .quad   0x4008000000000000 # 3.0
foo:
        movsd   xmm1, qword ptr [rip +
.LCPI0_0]
        addsd   xmm0, xmm1
        divsd   xmm0, xmm1
        ret
```

clang-21 -O2
https://godbolt.org/z/Tqvsv4Tqs

```
.LCPI0_0:
        .quad   0x4008000000000000 # 3.0
.LCPI0_1:
        .quad   0x3fd5555555555555 # 1/3
foo:
        addsd   xmm0, qword ptr [rip +
.LCPI0_0]
        mulsd   xmm0, qword ptr [rip +
.LCPI0_1]
        ret
```

clang-21 -O2 -freciprocal-math
https://godbolt.org/z/6sn7G4r4P

Fast math in other languages

- Different languages have different approach towards IEEE 754
- C Standard disallows fast-math optimizations (in Annex F)
  - Usually can be enabled with flags (e.g. -ffast-math)
  - Used e.g. in game engines
- Same applies for Swift language
- Java and C# allow to perform computations with higher precision than resulting type
- Go allows fusion of operations
  - Only if there are no explicit casts
- Fortran allows IEEE 754 violations to improve performance

Optimizations: Vectorization

- -fassociative-math and -fno-signed-zeros allow compiler to reorder loop iterations and vectorize loop

```c
#include <math.h>

float sum256(float *arr1,
             float *arr2) {
  float s;
  int i;
  s = 0.0f;
  for (i = 0; i < 256; i++) {
    s += arr1[i] * arr2[i];
  }
  return s;
}
```

```
.LBB0_1:
  movss   xmm1, dword ptr [rdi + 4*rax]
  mulss   xmm1, dword ptr [rsi + 4*rax]
  addss   xmm0, xmm1
  # Loop NOT vectorized
  inc     rax
  cmp     rax, 256
  jne     .LBB0_1
```

clang-21 -O2 -fno-unroll-loops
https://godbolt.org/z/v5vvEb1Th

```
.LBB0_1:
  movups  xmm1, xmmword ptr [rdi + 4*rax]
  movups  xmm2, xmmword ptr [rsi + 4*rax]
  mulps   xmm2, xmm1
  addps   xmm0, xmm2
  # Loop vectorized
  add     rax, 4
  cmp     rax, 256
  jne     .LBB0_1
```

clang-21 -O2 -fno-unroll-loops -fassociative-math -fno-signed-zeros
https://godbolt.org/z/j6P1ohcTr

Optimizations: Algebraic simplifications

- Fast-math may allow more simplifications or compile-time evaluation
- -fassociative-math + -fno-signed-zeros

```
double foo(double val) {
  return 3.0 + val - 3.0;
}
```

```
foo:
  addsd   xmm0, qword ptr [rip + .LCPI0_0] # +3.0
  addsd   xmm0, qword ptr [rip + .LCPI0_1] # -3.0
  ret
```

clang-21 -O2
https://godbolt.org/z/hjzWejv38

```
foo:
  ret
```

clang-21 -O2 -fassociative-math -fno-signed-zeros
https://godbolt.org/z/a5GME55MM

- -fassociative-math + -fno-signed-zeros + -freciprocal-math

```
double foo(double val) {
  return 3.0 * val / 3.0;
}
```

```
foo:
  movsd   xmm1, qword ptr [rip + .LCPI0_0]
  mulsd   xmm0, xmm1 # * 3.0
  divsd   xmm0, xmm1 # / 3.0
  ret
```

clang-21 -O2
https://godbolt.org/z/nancn6j8x

```
foo:
  ret
```

clang-21 -O2 -fassociative-math -fno-signed-zeros -freciprocal-math
https://godbolt.org/z/dMeKc8PW9

Optimizations: Fused instructions

- Some architectures support fusion of floating point instructions
  - E.g. Fused Multiply Add in x86-64
- Rounding of these instructions may be different from sequence of roundings of their primitive operations
- Enabled via -ffpcontract=on flag

```
double foo(double *val) {
  double acc = 1;
  for (int i = 0; i < 100; ++i) {
    acc = acc * val[i] + val[i + 1];
  }
  return acc;
}
```

```
.LBB0_1:
        mulsd   xmm0, xmm1 # multiply
        movsd   xmm1, qword ptr [rdi + 8*rax]
        addsd   xmm0, xmm1 # add
        inc     rax
        cmp     rax, 101
        jne     .LBB0_1
        ret
```

clang-21 -O2 -fno-unroll-loops
https://godbolt.org/z/qTTY4r7Yx

```
.LBB0_1:
        vmovsd  xmm2, qword ptr [rdi + 8*rax]
        vfmadd213sd     xmm0, xmm1, xmm2 # fused multiply-add
        inc     rax
        vmovapd xmm1, xmm2
        cmp     rax, 101
        jne     .LBB0_1
        ret
```

clang-21 -O2 -fno-unroll-loops -ffp-contract=on -march=haswell
https://godbolt.org/z/zvPTo9fz4

Optimizations: Finite only

- -ffinite-math-only enables optimizations that rely on lack of NaN or Inf values (and corresponding runtime exceptions)
- This allows to
  - Remove NaN and Inf checks
  - x == x => true
  - x / x => 1
- With -fno-signed-zeros:
  - x – x => 0
  - 0 / x => 0
  - x * 0 => 0

```
int foo(double val) {
  return val == val;
}
```

```
foo:
  ucomisd xmm0, xmm0
  setnp   al
  ret
```

clang-21 -O2

https://godbolt.org/z/v5YEq7hbY

```
foo:
  mov     al, 1
  ret
```

clang-21 -O2 -ffinite-math-only

https://godbolt.org/z/TYab9sndr

```
double foo(double val) {
  return val * 0;
}
```

```
foo:
  xorpd   xmm1, xmm1
  mulsd   xmm0, xmm1
  ret
```

clang-21 -O2

https://godbolt.org/z/3e8sM1dh9

```
foo:
  xorps   xmm0, xmm0
  ret
```

clang-21 -O2 -ffinite-math-only -fno-signed-zeros

https://godbolt.org/z/zoWrsfr1h

Optimizations: Errno

- -fno-math-errno flag allows compiler to ignore errno global variable effects in math functions
- In particular this enables better inlining and simplification of library functions
- In Rust standard math operations are compiled to LLVM intrinsics which do not set errno

```c
#include <math.h>

double foo(double x) {
  return sqrt(x);
}
```

```
foo:
        xorpd   xmm1, xmm1
        ucomisd xmm0, xmm1
        jb      sqrt@PLT
        sqrtsd  xmm0, xmm0
        ret
```

clang-21 -O2

```
foo:
        sqrtsd  xmm0, xmm0
        ret
```

clang-21 -O2 -fno-math-errno

```rust
pub fn foo(num: f64) -> f64 {
    num.sqrt()
}
```

```
foo:
        sqrtsd  xmm0, xmm0
        ret
```

Issues with fast-math: Checks

- Usage of Inf or NaN with -ffinite-math-only is UB
- Explicit checks for such values will be removed by compiler

```c
#include <math.h>

int foo(float f) {
  return isnan(f);
}
```

```
foo:
  xor     eax, eax
  ucomiss xmm0, xmm0
  setp    al
  ret
```

clang-21 -O2
https://godbolt.org/z/MPPoc5oea

```
foo:
  xor     eax, eax
  ret
```

clang-21 -O2 -ffinite-math-only
https://godbolt.org/z/nv33ffaqx

Issues with fast-math: Rounding

- Some algorithms use special order of computations to reduce rounding error
- With -fassociative-math compiler may ignore this order and “optimize” these algorithms as if there is no rounding
- E.g. Kahan algorithm

```
float sum(float *arr) {
  float s;
  int i;
  s = 0.0f;
  for (i = 0; i < 256; i++) {
    s = s + arr[i];
  }
  return s;
}
```

```
sum_kahan:
  xorps   xmm0, xmm0
  xor     eax, eax
.LBB0_1:
  addss   xmm0, dword ptr [rdi +
4*rax]
  inc     rax
  cmp     rax, 256
  jne     .LBB0_1
  ret
```

clang-21 -O2 -ffast-math -fno-unroll-loops -fno-vectorize
https://godbolt.org/z/rYebWsfco

```
float sum_kahan(float *arr) {
  float s, c, t, y;
  int i;
  s = c = 0.0f;
  for (i = 0; i < 256; i++) {
    y = arr[i] - c;
    t = s + y;
    c = (t - s) - y;
    s = t;
  }
  return s;
}
```

Rust approach

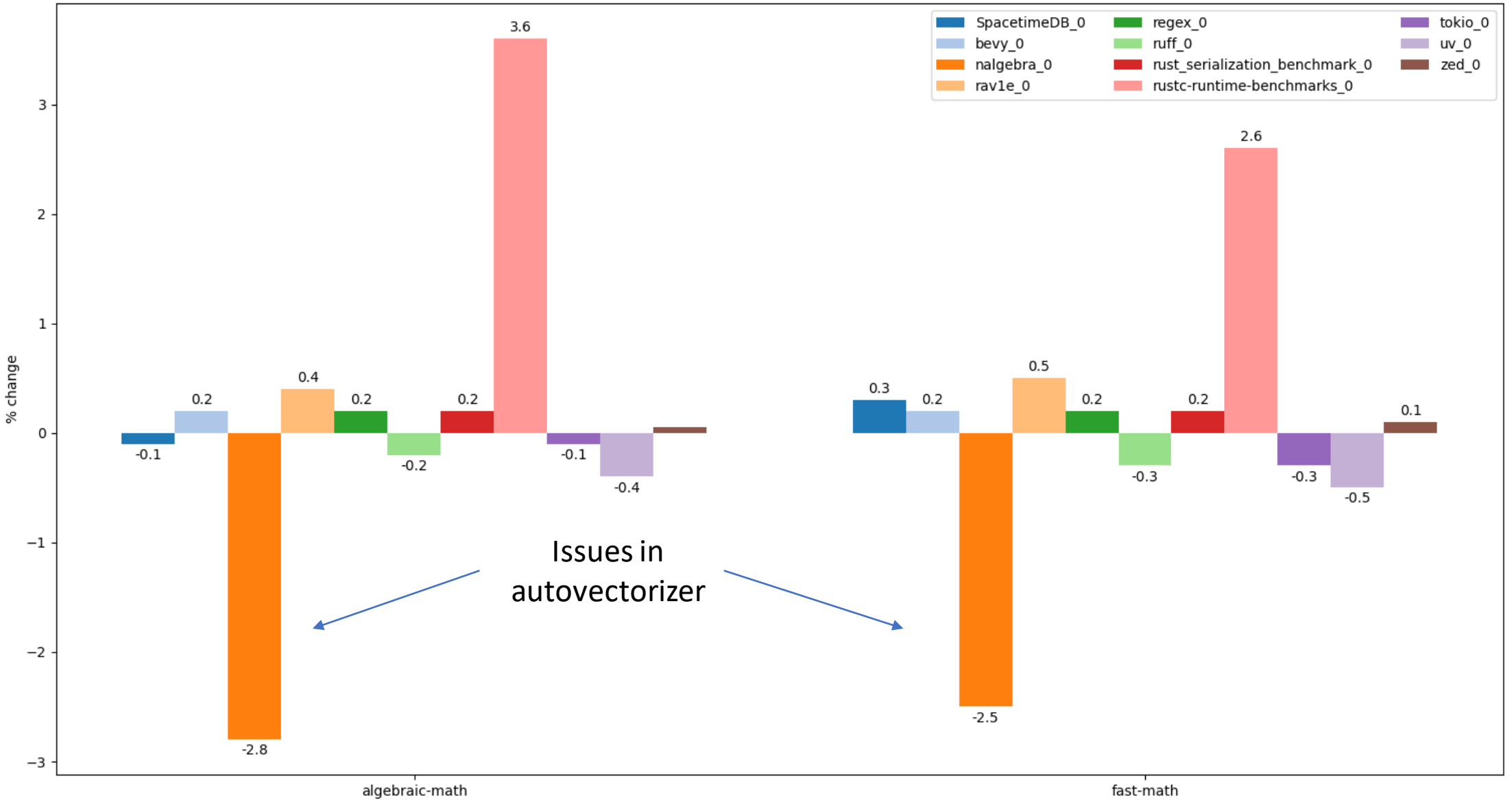
- Developers of Rust rejected to enable opportunities for global fast-math optimizations
- Due to potentially unsafe usage of these optimizations by users (especially in project dependencies)

> “Everything about this flag fundamentally clashes with the idea of robust, compositional system design. It was a terrible mistake to ever add it to C compilers, and Rust should not repeat that mistake. Rust should instead explore alternative, less fragile ways of exposing those semantics.” (Ralf Jung, rust #21690)

Workarounds: Intrinsics

- Rust provides dedicated intrinsics for enabling fast-math optimizations locally
  - std::intrinsics::f{op}_fast
  - std::intrinsics::f{op}_algebraic
  - f(16/32/64)::algebraic_{op}
- Disadvantages:
  - Currently require nightly (unstable) compiler build
  - Currently there are no wrapper types which hide usage of intrinsics
    - Authors of math libraries have to implement two versions of code (fast-math and normal)

Performance with fast-math



[Compiler branch]

[Compiler branch]

Conclusions

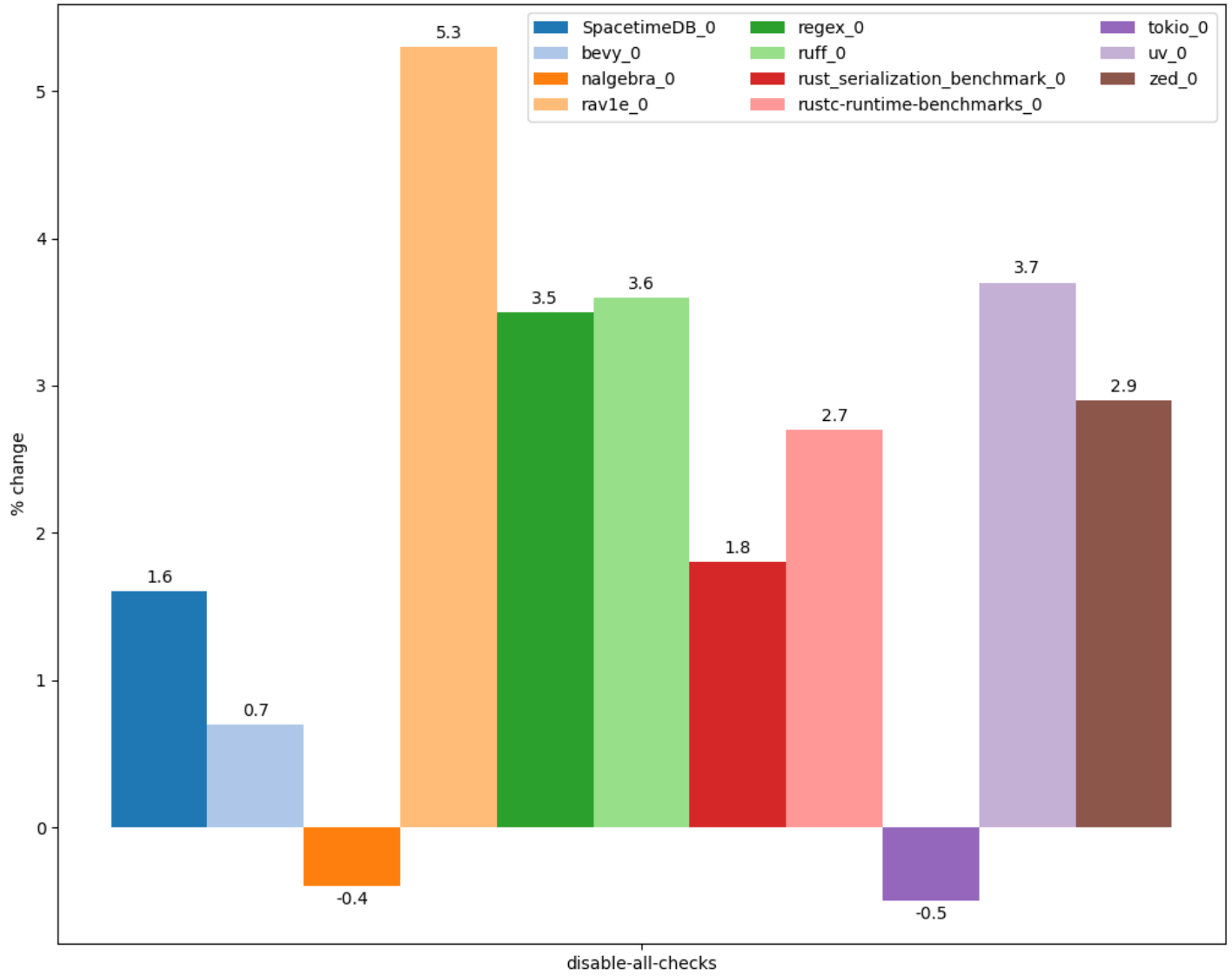
- Rust does not provide a way to enable fast-math optimizations globally
- Can be enabled for particular operations via intrinsics
  - Currently only in nightly toolchain
- Fast-math may both improve and degrade performance

Recommended readings

- Optimizations enabled by -ffast-math (Krister Walfridsson, GCC)
- Towards Useful Fast-Math (LLVM Dev Meeting 2024)
- Beware of fast-math
- P3715R0: Tightening floating-point semantics for C++
- Taming Floating-Point Sums

Conclusions

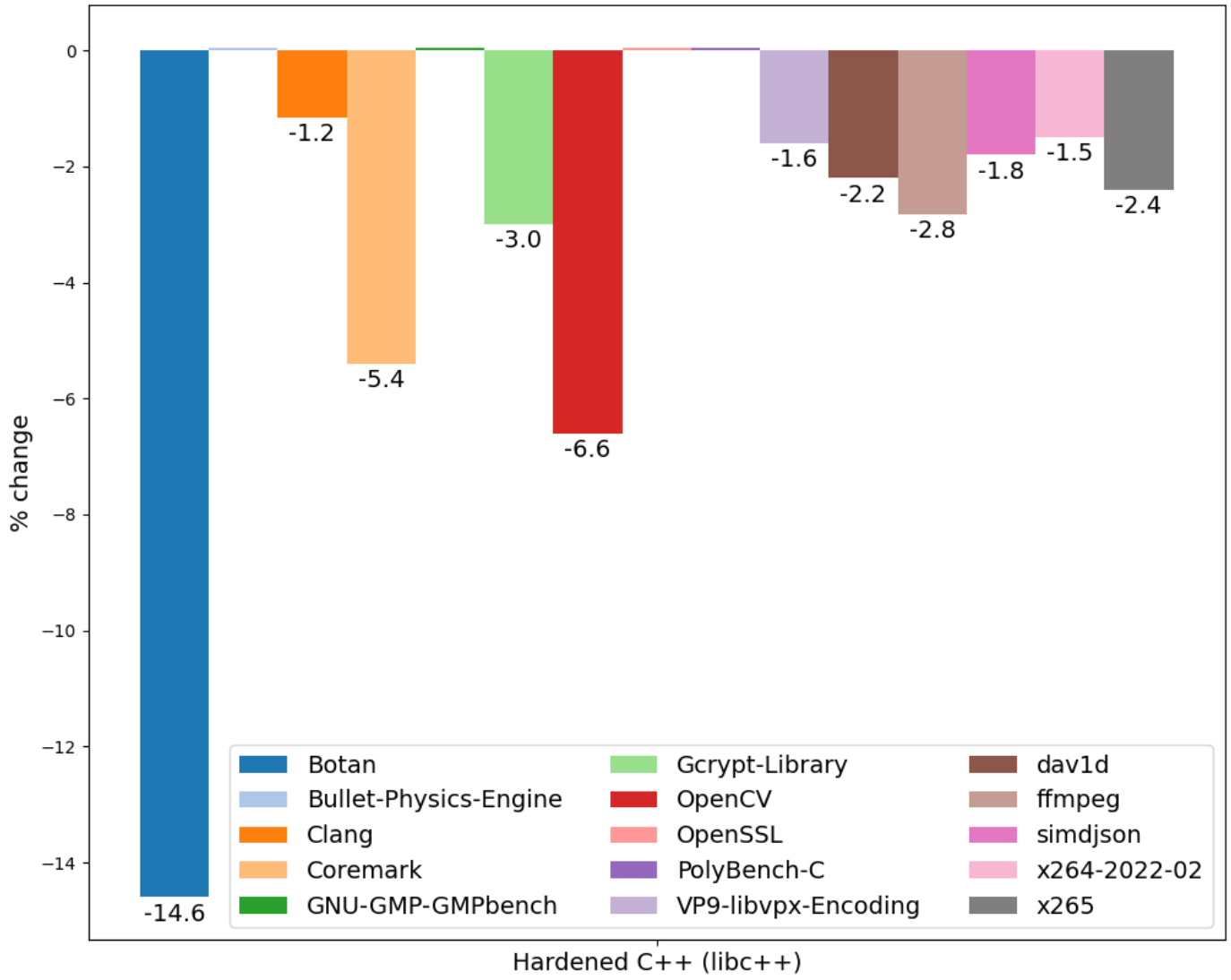
Speedup with disabling (most) Rust checks



- Disabled bounds and arithmetic checks, overflow checks in stdlib containers, UTF-8 string checks
- Not counting overheads for redundant initializations (expecting >= 1%)

[Compiler branch]

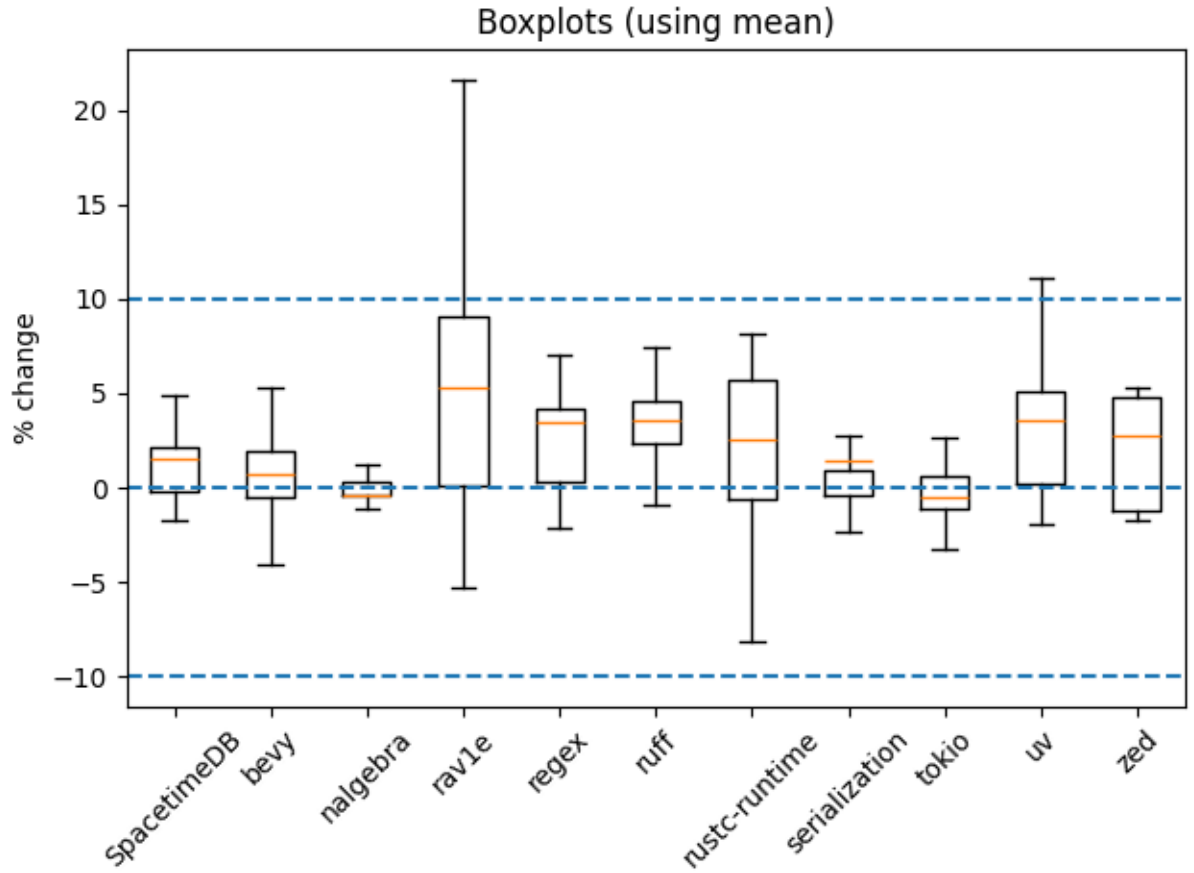
Slowdown in Hardened C++



- Enabled hardening:
  - -D_FORTIFY_SOURCE=3
  - -fsanitize=bounds,object-size
  - Hardened STL
  - (initialization not enabled to match Rust)
- Unlike Rust majority of benchmarks **have not been tuned** for Hardened C++ case

[Compiler branch]

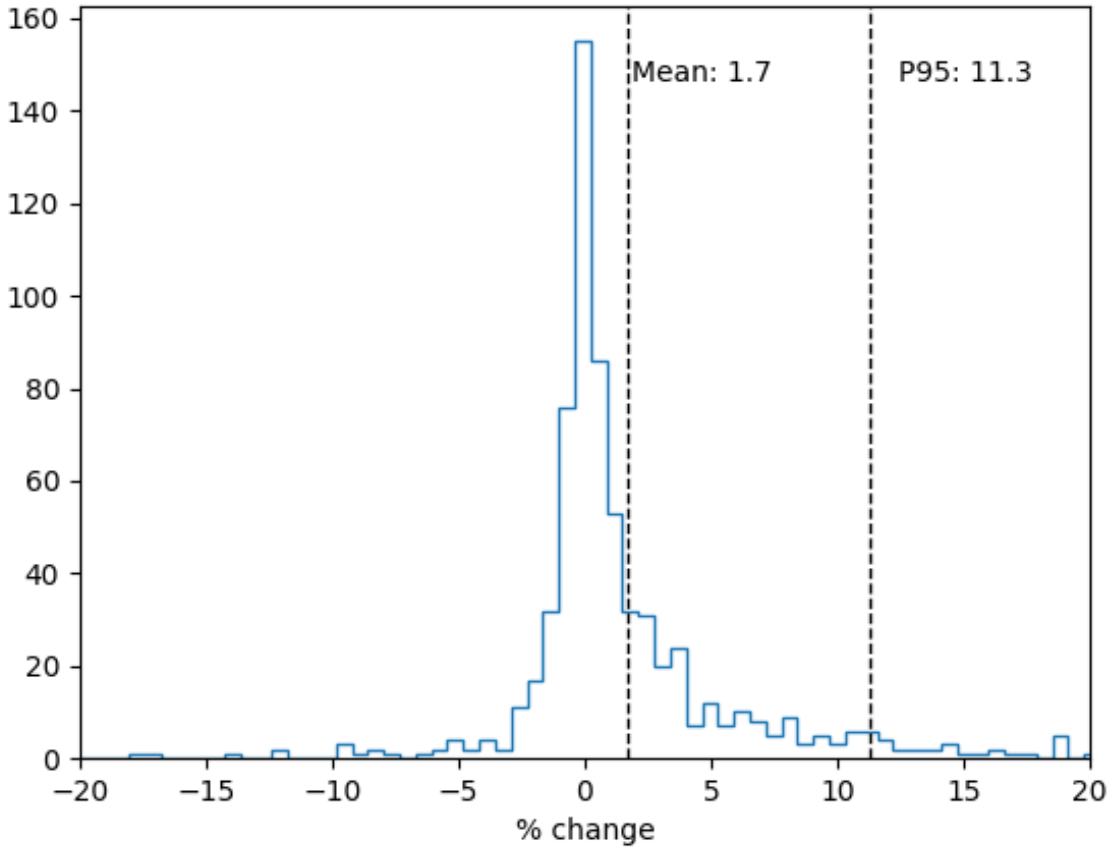
Distribution of overheads in benchmarks



- Distribution of all benches in each project
- Similar tests with different parameters were averaged
- Data is **biased**:
  - Scenarios which are covered by many benches dominate
- Expecting at least +1% mean due to forced initialization

[Plotting script]

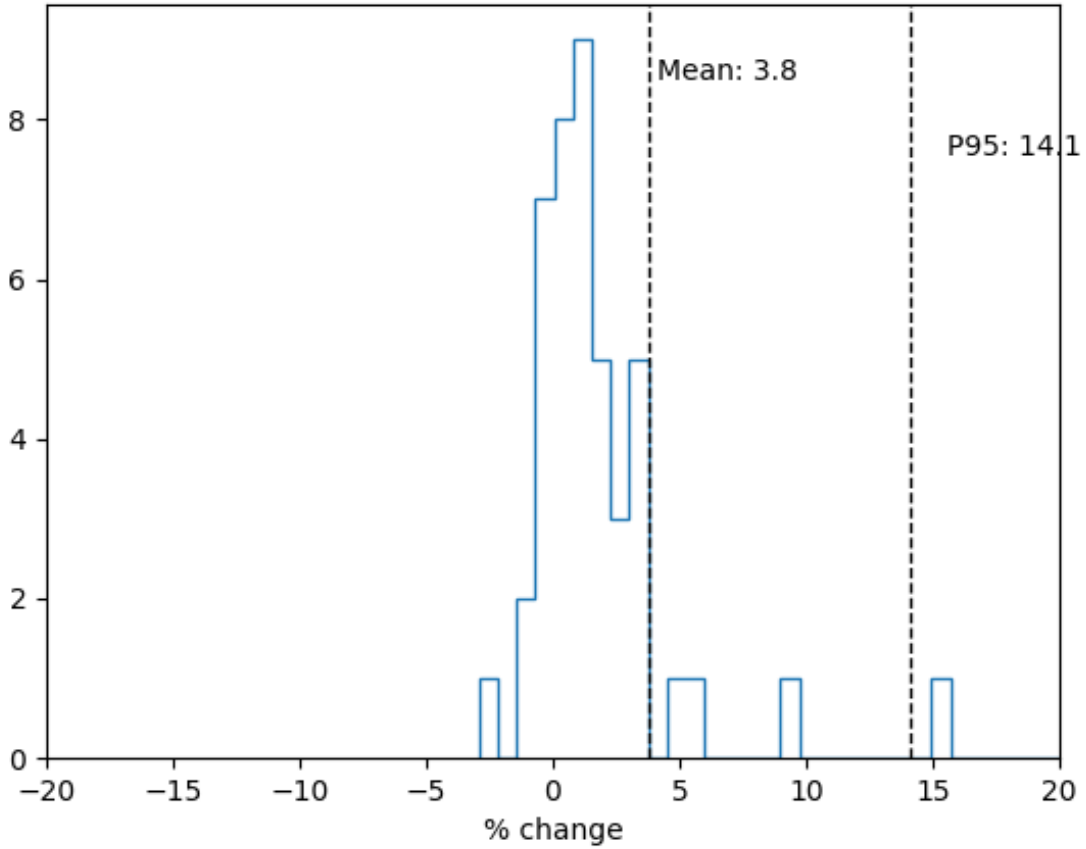
Distribution of overheads in benchmarks



- This graph is a pure illustration and does not prove anything!
- Distribution of all benches in all projects
- Similar tests with different parameters were averaged
- Data is **biased**:
  - Projects with larger numbers of benches dominate
  - Scenarios which are covered by many benches dominate
- Expecting at least +1% mean due to forced initialization

Plotting script

Distribution of overheads in Hardened C++ benchmarks



- This graph is a pure illustration and does not prove anything!
- Much fewer benchmarks than for Rust
- Unlike Rust majority of benchmarks **have not been tuned** for Hardened C++ case

Plotting script

Conclusions

- Rust community keeps balance between safety and performance:
  - Main focus is on memory safety guarantees
  - Other checks are enabled if they do not hurt performance
  - For example integer overflow errors are disabled by default
- Overhead of runtimes checks depends on the project
  - Usually under 5% on average (P95 11%) and comparable to Hardened C++
- Language features which affect performance the most:
  - Bounds checks, mandatory initialization, lack of inplace initialization
- These checks will most likely be further optimized in future versions of Rust and LLVM
- But their negative effects will never be completely avoided (at least in current language version)
- So they will always require workarounds (e.g. unsafe code) to reach optimal performance

3dman_eu, https://www.needpix.com/photo/1171460/weigh-plus-minus-horizontal-plus-minus-balance-sheet-cost-calculation-course

What we didn’t talk about …

- Codegen units
  - Similar to unity builds in C/C++
- Implementation of panics (exceptions)
  - Similar to C++ exceptions
- GC-sections by default
- ABI features
- Disabled PLT
- Implementation of virtual methods
- Stack Clashing checks

https://vectorportal.com/vector/timeglass/34811

The End

Many thanks to our kind reviewers:

- Vadim Petrochenkov
  - https://github.com/petrochenkov
- Alexander Chichigin
  - https://github.com/Gabriel-Fallen
- Jakub Beránek
  - https://github.com/kobzol
- Alexander Monakov
  - https://github.com/amonakov
  - https://codeberg.org/amonakov
- Per Vognsen
  - https://mastodon.social/@pervognsen

Latest version of this presentation: https://github.com/yugr/rust-slides/blob/main/EN.pdf

Appendix

Preparation of this talk

- We collected and analyzed over 350 papers, blogposts and discussions about the topic (see materials)
- Overall time spent on preparation (Apr 2026):
  - > 3 man-months
- Analysis was conducted for
  - Rust 1.87.0 (May 2025)
  - Clang llvmorg-20.1.7 (June 2025)

Rust benchmarks

- Used standard Criterion benchmarks in each project (cargo bench), averaged with geomean
  - Some especially slow tests in oxipng were disabled
  - To counteract system noise we took minimum of average from several runs
  - Note that 1% geomean may mean that
    - All tests in project slowed down by 1%
    - 10% of tests slowed down by 10% (more likely)
  - Some tests could slow down (or speedup) by more than 10%
- Geomeans are not ideal
  - Biased by scenarios with more benches
- Further details here

C++ benchmarks

- Used Clang llvmorg-20.1.7
- Llvm-bench (benchmarks/cpp/llvm-bench)
  - Measured median time of -O2 compilation of CGBuiltin.cpp (the largest and slowest-to-compile file in LLVM) with Clang
  - Tested Clang was built with itself
- Ffmpeg-bench (benchmarks/cpp/ffmpeg-bench)
  - Based on Stack Smashing Protection and Its Performance Impact
  - Measured median time to convert FLV to MP4
- Phoronix Test Suite (benchmarks/cpp/PTS)
  - Popular benchmark suite from phoronix-test-suite.com
  - Measured botan, bullet, coremark, c-ray, dav1d, gcrypt, gmpbench, luajit, nginx, opencv, openssl, polybench-c, povray, rocksdb, simdjson, vpxenc, x265 (see #55)
  - Used median time of benchmark
  - Ignored tests with Stdev > 0.5% or changes < 1%
- To counteract system noise we took minimum of average from several runs

Hardware

- All benchmarks (except no-static) were run on Intel Xeon E-2236
  - With fixed 2 GHz frequency
  - With Ubuntu 24.04 OS
- Servers have been set up according to uInit methodology

TODO

- (Yugr) Benchmark meilisearch
- (Zak) Fast-math: update analysis (P3715, -fno-math-errno and -fno-trapping-math)
- (Zak) Struct reorder: fill analysis